

Title page

Title. Narrative Embeddings Do Not Outperform Typed Features for a Treatment-Switch Proxy of Treatment-Resistant Depression: Retrospective Cohort Study

Short title. Embeddings and Typed Features for TRD Prediction

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Abstract

Background. As many as 1 in 3 patients with major depressive disorder (MDD) progress to treatment-resistant depression (TRD) after failing successive antidepressant trials [1]. An electronic health record (EHR) can be given to a model as a typed feature vector, or written out as text and encoded by a pretrained transformer, and the two have not been compared. A third approach predicts a patient from the outcomes of their closest analogues, the premise behind clinical digital twins.

Objective. Two questions on one cohort. Whether a generalized pre-trained transformer embedding of a patient narrative outperforms a typed feature vector for predicting a treatment-switch-defined EHR proxy for incident TRD, and whether nearest-neighbor retrieval over that embedding predicts as well as a trained classifier.

Methods. Retrospective cohort of 42,579 adults with unipolar MDD from one community health system. Following Forthman et al [2], TRD was 3 or more antidepressant treatments within 1 year of the index date, the earliest antidepressant prescription after a documented depression diagnosis; predictors came only from the 730 days before it. Two representations were built from the same hand-picked fields: a typed feature vector, and a Markdown narrative written by fixed rules with no generative model, encoded by a pretrained sentence-transformer (4 encoders; primary Qwen3-Embedding-8B). Four classifiers were tuned by cross-validated grid search and scored on one held-out stratified 20% split (8,516 patients, 1,491 TRD-positive). A retrieval predictor scored each patient as the weighted mean TRD rate of 50 retrieved training patients under 4 retrieval schemes. A semantic-feature ablation shuffled each narrative concept across patients in turn to locate the embedded signal. Reporting follows TRIPOD+AI [3].

Results. The embedding did not outperform the feature vector: ROC AUC 0.657 (95% CI 0.643–0.672) against 0.649 (0.634–0.664), paired difference +0.008 (−0.003 to +0.019). Holding the classifier fixed, it helped logistic regression (+0.028) and hurt all 3 tree ensembles (−0.013 to −0.022). Discrimination stayed within 0.012 across 4 encoders. The ablation localized the signal to psychiatric history, medication burden and prior treatment exposure rather than to sociodemographic fields. Retrieval was informative, not competitive: nearest 0.594 (0.578–0.610) against random 0.499 on all 4 encoders, yet 0.063 short of the trained classifier on intervals that do not overlap.

Conclusions. Embedding matched, but did not exceed, a feature vector built from the same fields, and retrieval over that embedding lost to a fitted model. Both arms encode the same hand-picked inventory, so neither result shows that embedding removes the feature engineering. Discrimination near 0.65 is too modest for clinical use, and the outcome is a treatment-switch phenotype rather than verified non-response.

Keywords. treatment-resistant depression; major depressive disorder; electronic health records; clinical prediction model; machine learning; text embeddings; nearest-neighbor retrieval; feature engineering; TRIPOD+AI.

Introduction

Major depressive disorder is among the most prevalent and disabling medical conditions, and antidepressant pharmacotherapy is its first-line treatment. A substantial minority of patients fail to achieve remission despite successive adequate trials. These patients are described as having treatment-resistant depression (TRD), most commonly operationalized as the failure of at least two antidepressant trials of adequate dose and duration [4]. The sequential-treatment evidence base, anchored by the STAR*D program, established that remission rates fall sharply with each successive treatment step. Patients who move into a third or fourth step face a markedly lower probability of recovery [5]. TRD accordingly accounts for a disproportionate share of the clinical morbidity, functional impairment, and health-care cost attributable to depression [1].

Escalation options become both more necessary and less likely to succeed as resistance develops. There is therefore longstanding interest in identifying, as early as possible, which patients are at high risk of progressing to TRD. If such patients could be flagged at the point of their index antidepressant prescription, clinicians could in principle monitor them more closely and escalate sooner.

An electronic health record (EHR) does not say whether a patient got better, and rarely says whether a drug was given long enough at a high enough dose to count as a fair trial. What it does record is which antidepressants were prescribed and when. So studies define TRD from the prescription sequence itself, which is what makes a cohort of tens of thousands possible at all. A sequence is not proof of failure. A patient may be switched because the drug did not work, or because it was not tolerated, or because they stopped taking it. And the rule is not fixed: research groups count treatments, minimum durations, switches and augmentations differently, and changing the rule changes both how many patients count as TRD and which ones [6-8].

Others have already tried to predict TRD from what an EHR records, and the numbers they report vary far more than the methods do. One study of 35,246 adults reported a ROC AUC of 0.83 [9]. A model that was validated on a second health system reached 0.652 [10]. Adding clinical notes to structured fields reached 0.728 at a single site [11]. And in a three-system study, figures of 0.58 to 0.64 inside the site that built the model fell to 0.51 to 0.58 when it was tested elsewhere [12]. The spread comes from how each study defined TRD, how far ahead it predicted, and whether it was ever tested outside the site it was built in, so there is no single figure to beat. Two things survive the variation. Structured EHR fields carry real signal, and the sharpest

drops come at the point where a model meets a health system it was not built on [8,10-12].

Two questions about how to use such a record motivate this study. The first is representational. Predicting clinical outcomes with large, purpose-built EHR foundation models requires data and compute at scales most groups cannot reach [13,14]. Serializing an EHR into text and applying general-purpose language-model embeddings is a computationally efficient alternative that has proved competitive across clinical prediction tasks [15]. It has not been compared directly against a transparent, hand-crafted feature vector for TRD prediction. The comparison matters because the two are not equally auditable. A feature vector can be read, and a 4,096-dimensional embedding cannot, so a gain in discrimination is the only thing that would pay for the loss of interpretability.

The second question concerns prediction by analogy. A recurring proposal in precision psychiatry is that a patient's likely course can be read off the recorded courses of the patients most similar to them, an idea usually described as a clinical digital twin. An embedding makes that proposal directly testable, because similarity in the embedding space is a defined quantity and the neighbors it selects have observed outcomes. If the premise holds, retrieving a patient's closest analogues and averaging their outcomes should predict well. Whether it does, and how it compares with fitting a model to the same representation, has not been established for TRD.

One thing is true of both questions and bounds both answers, so we state it here rather than leaving it to be inferred. **The patient data were hand-picked in both arms.** Serializing a record and embedding it is often read as a way to stop choosing features, and that is not what happens here. Neither answer below is evidence about what a model would do with a raw record, because neither arm was given one.

Embedding a whole untailored record would be the interesting version of that, and this study does not attempt it. The obstacle is input length rather than principle.

This study evaluated whether EHR-derived patient data can predict this prescription-sequence definition of TRD within one year, and addressed the two objectives in order. To ensure methodological transparency and facilitate rigorous appraisal, we report it in full accordance with the TRIPOD+AI reporting guideline for clinical prediction models developed with regression or machine-learning methods [3]. First, we compared a transparent typed feature vector against generalized pretrained transformer embeddings of patient narratives, evaluating both with four standard classifiers on an iden-

tical cohort and held-out split. We repeated that comparison across four encoders and located the embedded signal with a semantic-feature ablation that permutes individual clinical concepts. Second, we tested nearest-neighbor retrieval over the same embedding against a random-retrieval control, and against the trained classifiers themselves. Field by field, what each representation receives is documented in Supplement S8.

Methods

Study design and data source

We conducted a retrospective cohort study using a frozen, de-identified extract from the Epic EHR of Saint Francis Health System, a community health system in Tulsa, Oklahoma. The extract included person, encounter, diagnosis, medication, procedure, laboratory/flowsheet, and medication-to-RxNorm mapping data. Patient and encounter identifiers were replaced with one-way hashes before transfer, and no direct identifiers or dates of birth were provided.

The delivered extract was case enriched rather than a census of the health system. All patients with a depression code on the problem list were retained, together with a random sample of patients without that flag at an approximate 4:1 ratio. The extract contained 501,718 patients, including 100,420 (20.0%) with and 401,298 (80.0%) without a problem-list depression flag. Eligibility for the analytic cohort was determined from encounter-level diagnoses rather than the problem-list sampling flag. Consequently, 12,530 of the 42,579 eligible patients (29.4%) entered through the randomly sampled group. Cohort proportions therefore describe this enriched sample and should not be interpreted as health-system prevalence estimates. The head-to-head comparison of representations is unaffected, both being derived from the same patients and scored on the same held-out split.

All analyses used data version DV260629v1 and patient version PV260710v1. Source files were cut on June 29, 2026, and analysis-ready tables were created on July 10, 2026. No subsequent data refresh entered the study. Patient index dates spanned 2016-2024. The study used a random internal validation split from the same source and period, not temporal or external validation. Additional information on the source tables, setting, sampling frame, diagnosis code sets, and data versions is provided in Supplement M1-M2.

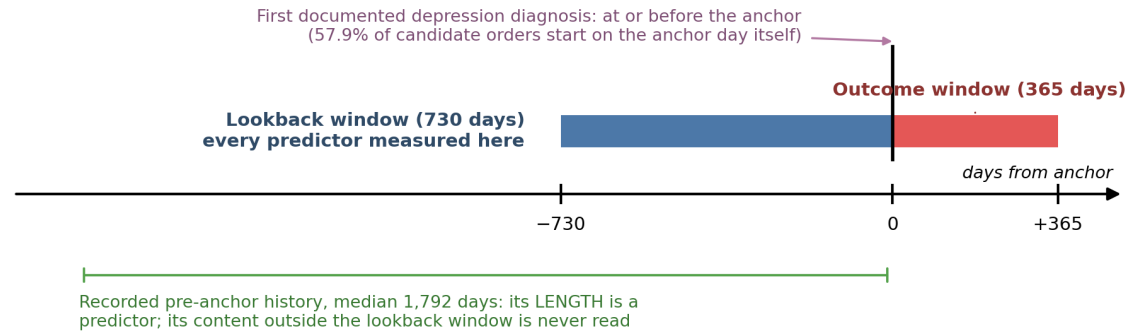
Participants, index date, and temporal windows

Eligibility required a qualifying MDD diagnosis, an eligible antidepressant index prescription, and that diagnosis recorded on or before that prescription. Patients with a bipolar disorder or schizophrenia-spectrum diagnosis were excluded. Each patient also needed at least 730 days of recorded EHR history before the index prescription and at least 365 days of follow-up after it. Those are the windows in which predictors and the outcome are measured. The cascade and the rejection reason at each step are reported in Supplement M2 (Table M1). These criteria yielded 42,579 patients.

The index date was the start date of each patient's earliest antidepressant prescription recorded on or after the first documented depression diagnosis. Each patient contributed one index date. Nothing in that selection rule looks forward. Dose, duration, response, and subsequent treatment changes play no part in choosing the index prescription, so the prediction point is the prescription date itself. Predictors were computed only from the 730-day lookback window ending on the index date, which Figure 1 draws. Post-index data entered in 2 places, confirming 365 days of follow-up and ascertaining the outcome. The lookback window and the outcome window therefore never overlap.

The index prescription is not the patient's first antidepressant, and not their first adequate trial. No adequacy, dose, or duration criterion enters its selection, and Supplement M2 gives the rules in full. It is the first antidepressant the record links to a documented depression diagnosis. Patients with prior antidepressant exposure were therefore kept rather than excluded, and that exposure was measured in the lookback window and retained as a predictor.

Anchor (time zero): the earliest antidepressant prescription recorded on or after the first documented depression diagnosis. Prediction is made here.



Eligibility requires at least 365 days of post-anchor follow-up. Three or more antidepressant treatments inside the outcome window label a patient TRD-positive. No property of the anchor, and no predictor, reads data recorded after the anchor date.

Figure 1. Time zero, the lookback window, and the outcome ascertainment window. The index is the earliest antidepressant prescription recorded on or after the patient's first documented depression diagnosis. Nothing that happens after that date is used to choose it, so the prediction point is the prescription date. Every predictor is measured inside the 730-day lookback window. Recorded history of ten extends further back, with a median of 1,792 days, and its length is itself a predictor. Its content outside the lookback window is never read. Post-index data enters in 2 places only: eligibility requires at least 365 days of follow-up, and the outcome is ascertained over those 365 days.

Outcome

The prediction target was a predefined EHR proxy for incident TRD, generated upstream and used without modification. Consistent with the operational definition used by Forthman et al [2], patients with MDD were classified as TRD positive if they received at least 3 distinct antidepressant treatments during the 365 days after the index date, including the index agent as the first treatment. Equivalently, the outcome required at least 2 post-index antidepressant changes. A treatment was a distinct antidepressant agent rather than a prescribing event. Augmentation without replacement and restarting a previously used agent did not increment the count, and augmentation enters the study as a pre-index predictor instead.

This outcome is a treatment-switch proxy rather than direct evidence of inadequate response to 2 adequate trials [4,5]. Symptom response

was not systematically available, and dose and duration adequacy were not verified for the counted treatments. Switching may therefore reflect nonresponse, intolerance, adverse effects, cost, insurance constraints, fragmented care, patient preference, or prescriber practice, and it depends on continuity of care and access, which differ across demographic groups [16,17]. The label was not validated against chart review or standardized symptom measures. These inferential boundaries and proposed validation analyses are detailed in Supplement M3, and their consequences for the study's claims in the Discussion, *Limitations*.

Predictors and patient representations

Predictors were specified a priori on clinical and prior-literature grounds before model fitting or inspection of outcome associations. Domains included depression characteristics, psychiatric and substance-use comorbidity, medical comorbidity, prior antidepressant exposure and medication burden, prescribing constraints, health care utilization, and sociodemographic and social-determinant variables [18-21]. Selection rationale is given in Supplement M4 and the complete inventory in Multimedia Appendix 1. Only structured EHR fields available within the pre-index window were used.

Both representations were built from the same curated field inventory, and neither was built from a raw record. Predictor selection happened once, before either representation existed, and both then encoded that selection. This bounds every comparison below: what is compared is two encodings of one tailored record, not a tailored record against an untailored one. Each patient's timeline-sliced record was converted independently into 2 representations. The typed feature representation (FEATURE) assigned explicit data types to continuous variables, binary and multilabel indicators, and nominal categorical variables. Continuous variables included age, utilization and duration measures, medication burden, and counts of adequate antidepressant trials, with an adequate trial defined as at least 42 days of continuous exposure to an agent. After categorical expansion and exclusion of 3 vital-sign variables for missingness, the FEATURE representation expanded to 92 columns.

For the embedded representation (EMBEDDED), the same pre-index record was deterministically rendered as a human-readable Markdown narrative and encoded as a fixed-length dense vector. The renderer walks a fixed template of section headings and field labels, in a fixed order, over the fields named above. Absent findings are printed explicitly and unrecorded ones as a literal token, so every

patient reaches the encoder through the same constant form. Narrative construction used predefined rules and no generative model, ensuring that every statement was traceable to the structured record. That guarantee is worth the loss of fluency, because a generated summary introduces statements the record does not support at measured rates [22] and here the narrative *is* the predictor. Four pretrained sentence-transformer encoders were evaluated independently: bge-small-en-v1.5 [23], bge-en-icl [24], Qwen3-Embedding-4B, and Qwen3-Embedding-8B [25], with Qwen3-Embedding-8B designated as the primary encoder.

Predictor selection was identical for the 2 representations, and both were built from the same pre-index slice of the record. What differed was how some of those selected fields were encoded. Supplement S8 is a field-level crosswalk that sets the 2 encodings side by side across 28 source fields. For 17 fields the 2 sides carry the same information. The remaining 11 are asymmetric, meaning that one representation is told something the other is not.

The narrative holds more in 10 of those 11 fields. Some fields it renders have no column in the feature vector at all. Others it prints as the raw Epic value where the feature vector collapses that value into coarse categories. Supplement M5 states the full encoding rules and Supplement S4 reproduces 2 example narratives. The eleventh field runs the other way, recorded history length, which the narrative does not render. A head-to-head result therefore compares 2 whole pipelines rather than data format alone, and the crosswalk is what makes that bound checkable field by field.

Missing data and sample partition

No statistical imputation was performed. Mean body mass index and mean systolic and diastolic blood pressure were excluded because of substantial and outcome-associated missingness. Missing categorical values were retained as an explicit category in FEATURE and as the literal token “Missing” in EMBEDDED. No missing continuous values reached the estimators. Detailed missingness frequencies and rationale are provided in Supplement M6.

We created one stratified 80:20 train-test split and used it for every evaluation arm. The training set comprised 34,063 patients, including 5,964 TRD-positive patients (17.5%). The held-out test set comprised 8,516 patients, including 1,491 TRD-positive patients (17.5%). Across 100 predictor rows, the maximum absolute standardized mean difference between the 2 sets was 0.036. For

retrieval analyses, only training patients were placed in the searchable neighbor pool. Test patients served solely as query anchors, which prevents self-retrieval and any use of a test outcome as a neighbor label. Events-per-variable calculations and further leakage safeguards appear in Supplement M7-M8.

Model development

Standard machine learning. Logistic regression, random forest, gradient boosting, and XGBoost classifiers were trained separately on each representation. For FEATURE, a column transformer standardized continuous variables, one-hot encoded categorical variables while ignoring unseen levels at inference, and cast Boolean variables to integers. EMBEDDED was processed as an all-numeric block. Preprocessing and estimation were combined within a single pipeline and tuned using 5-fold cross-validated grid search in the training set, optimizing area under the receiver operating characteristic curve (ROC AUC). On every fold, fitted preprocessing parameters and category vocabularies were estimated from that fold's training rows only. The best cross-validated estimator was refit on the full training set and generated probabilities for the untouched test set (Supplement M9).

Neighbor-weighted retrieval prediction. For EMBEDDED only, we also estimated risk without fitting a model, as the weighted mean of TRD labels among $K = 50$ retrieved training-set neighbors. This tests the digital-twin premise directly: if a patient's course can be read from the courses of their closest analogues, retrieval should predict well. Retrieval schemes selected the nearest or the random training patients, with random retrieval as the negative control. Weighting was uniform or based on anchor-neighbor cosine similarity. Detailed equations, retrieval controls, and the evaluation grid are provided in Supplement M10 and M12.

Semantic-feature ablation. To estimate direct reliance of the embedded pipeline on selected concepts, we permuted 1 narrative section or field at a time across patients, re-embedded the perturbed narratives, and applied the already-trained classifiers without retraining. Permutation preserved each concept's marginal distribution while severing its patient-level association with outcome. The tested concepts were psychiatric history, medication burden, prior treatment exposure, treatment contraindications, race/ethnicity, and social determinants of health. Change in ROC AUC relative to the unperturbed model quantified direct-input reliance. This analysis was not designed to establish fairness, rule out reconstruction of a permuted concept from correlated fields, or evaluate subgroup cali-

bration. Specifications, including what each concept’s permutation destroys, are provided in Supplement M11.

Statistical analysis and performance metrics

Performance in the held-out test set was characterized using ROC AUC, calibration slope and intercept, and sensitivity, specificity and the positive and negative likelihood ratios at the threshold maximizing the Youden J statistic. That threshold was selected and evaluated in the same test set, so operating-point estimates are descriptive and optimistic rather than deployable clinical thresholds. Calibration estimates apply to the case-enriched cohort’s outcome frequency and were not recalibrated to population prevalence.

An interval containing zero means no difference was detected, not that the two representations are equivalent. No equivalence or non-inferiority margin was set in advance, so this paper makes no equivalence claim anywhere.

Confidence intervals come from 1,000 nonparametric bootstrap resamples of the test-set patients, detailed in Supplement M13. The subgroup analysis computes 240 between-group contrasts, so its P values are adjusted across that whole set by the Benjamini-Hochberg procedure, controlling the false discovery rate at 5%.

Comparisons between two prediction vectors are paired. Each bootstrap draw scores both prediction vectors on the same resampled patients before their difference in ROC AUC is taken, which keeps the patient draw out of the comparison. Classifier-matched contrasts hold the learner fixed and vary only the representation. The best-feature-versus-best-embedded comparison was chosen after the results were in, so it is descriptive rather than a test. Retrieval against the trained classifiers is reported as marginal rather than paired intervals. Pairing would narrow those intervals, so reporting them unpaired is the conservative choice.

Reproducibility and software

All random processes were seeded. Analyses were implemented in Python using scikit-learn [26], XGBoost [27], and sentence-transformer encoders [28]. The analysis code is publicly available [29].

Results

The results are reported in the order of the two objectives. Participant flow and cohort composition come first, then the comparison between the two representations, then the retrieval predictor, and last the checks that bear on both.

Participant flow

Of the 501,718 patients in the delivered extract, 42,579 satisfied every inclusion and exclusion criterion. Supplement M2 (Table M1) counts the rejections at each stage of the hierarchical cascade. Those 42,579 are the analysis cohort throughout, and 7,455 of them (17.5%) were TRD-positive.

Cohort characteristics

The cohort was middle-aged (median age 55 years, IQR 38–70), predominantly female (72.5%), White/Caucasian (80.0%) and English-speaking (98.9%). Table 1 gives every selected characteristic, overall and by TRD status, and Supplement S9 gives within-subgroup TRD prevalence. The characteristics most associated with TRD were a flagged suicidality history, severe MDD coding, and a broad band of psychiatric and substance-use comorbidity, which is the same profile the models later rely on.

Table 1. Selected cohort characteristics by TRD status. Continuous variables are median (IQR); categorical/boolean entries are *n* (%). SMD = standardized mean difference (TRD-positive vs TRD-negative).

Characteristic	Level	Overall	TRD+	TRD–	SMD
Demographics					
Age (years)	median (IQR)	55 (38–70)	51 (36–66)	56 (39–71)	–0.193
Age band	18–29	5,513 (12.9%)	1,135 (15.2%)	4,378 (12.5%)	0.080
	30–44	9,025 (21.2%)	1,836 (24.6%)	7,189 (20.5%)	0.100
	45–64	13,268 (31.2%)	2,445 (32.8%)	10,823 (30.8%)	0.043
	65+	14,773 (34.7%)	2,039 (27.4%)	12,734 (36.3%)	–0.192

Characteristic	Level	Overall	TRD+	TRD–	SMD
Sex	Female	30,850 (72.5%)	5,552 (74.5%)	25,298 (72.0%)	0.055
Race/ethnicity	White/Caucasian	24,079 (80.0%)	5,942 (79.7%)	28,137 (80.1%)	–0.010
	Black/African American	2,368 (5.6%)	401 (5.4%)	1,967 (5.6%)	–0.010
	Am. Indian/Alaska Native	2,033 (4.8%)	394 (5.3%)	1,639 (4.7%)	0.028
	Missing	297 (0.7%)	46 (0.6%)	251 (0.7%)	–0.012
Depression phenotype					
MDD recurrence	Recurrent	11,213 (26.3%)	2,189 (29.4%)	9,024 (25.7%)	0.082
	Single episode	28,206 (66.2%)	4,721 (63.3%)	23,485 (66.9%)	–0.074
MDD severity	Severe	2,286 (5.4%)	720 (9.7%)	1,566 (4.5%)	0.204
	Moderate	7,233 (17.0%)	1,300 (17.4%)	5,933 (16.9%)	0.014
	Psychotic	112 (0.3%)	35 (0.5%)	77 (0.2%)	0.043
	Unspecified	28,951 (68.0%)	4,803 (64.4%)	24,148 (68.8%)	–0.092
Psychiatric and substance comorbidity					
Suicidality flagged	True	1,753 (4.1%)	641 (8.6%)	1,112 (3.2%)	0.232
Anxiety disorder	True	22,401 (52.6%)	4,499 (60.3%)	17,902 (51.0%)	0.190
Substance use disorder (any)	True	9,265 (21.8%)	2,006 (26.9%)	7,259 (20.7%)	0.147

Characteristic	Level	Overall	TRD+	TRD-	SMD
Insomnia	True	9,350 (22.0%)	2,021 (27.1%)	7,329 (20.9%)	0.147
PTSD	True	1,541 (3.6%)	450 (6.0%)	1,091 (3.1%)	0.141
Alcohol use disorder	True	1,875 (4.4%)	501 (6.7%)	1,374 (3.9%)	0.125
Opioid use disorder	True	1,281 (3.0%)	337 (4.5%)	944 (2.7%)	0.098
Adjustment disorder	True	2,918 (6.9%)	663 (8.9%)	2,255 (6.4%)	0.093
Medical comorbidity					
High cholesterol	True	15,747 (37.0%)	2,202 (29.5%)	13,545 (38.6%)	-0.191
Uncontrolled hypertension	True	15,680 (36.8%)	2,366 (31.7%)	13,314 (37.9%)	-0.130
Treatment and utilization					
Active med count	median (IQR)	1 (0-2)	1 (0-2)	1 (0-2)	0.084
Encounter count	median (IQR)	21 (8-46)	17 (6-40)	21 (8-47)	-0.140
Pre-index history (days)	median (IQR)	1,792 (1,216-2,546)	1,629 (1,141-2,335)	1,830 (1,239-2,587)	-0.198
Any prior treatment exposure recorded	True	10,230 (24.0%)	1,793 (24.1%)	8,437 (24.0%)	0.001

Characteristic	Level	Overall	TRD+	TRD−	SMD
Prior adequate AD trial (any class)	True	6,276 (14.7%)	1,050 (14.1%)	5,226 (14.9%)	−0.023
Benzodiazepine days recorded	True	4,724 (11.1%)	989 (13.3%)	3,735 (10.6%)	0.081
Hypnotic recorded	True	1,661 (3.9%)	292 (3.9%)	1,369 (3.9%)	0.001
Augmentative therapy used	True	422 (1.0%)	108 (1.4%)	314 (0.9%)	0.052

Model discrimination

The embedding did not outperform the feature vector. Embedded logistic regression achieved the highest discrimination on the embedded representation (ROC AUC 0.657, 95% CI 0.643–0.672), and XGBoost led the feature-vector models (0.649, 95% CI 0.634–0.664). All eight representation-by-classifier combinations appear in Figure 2A and Table 2. The best embedded model scored 0.008 higher than the best feature-vector model, on an interval from −0.003 to +0.019 that crosses zero (Figure 2B). The comparison is paired: every bootstrap draw scores both models on the same patients, so the interval measures the gap between models rather than the luck of the draw. An embedded gain larger than +0.019 ROC AUC falls outside it. Discrimination was modest throughout, from 0.623 at the lowest configuration to 0.657 at the highest, against a 17.5% TRD-positive rate.

Which representation won depended on the learner. Holding the classifier fixed, the embedding helped logistic regression (+0.028, 95% CI +0.017 to +0.039). It hurt all three tree ensembles: random forest by 0.022 (95% CI 0.011 to 0.033), gradient boosting by 0.013 (0.001 to 0.026), and XGBoost by 0.013 (0.002 to 0.025). Tuning grids did not differ between representations, so this is a property of the representation rather than of its tuning.

Table 2. Discrimination of the four classifiers on each representation (held-out test set). 95% CIs are bootstrap percentile intervals. FEATURE-side values are from the 92-column feature vector.

Representation	Classifier	ROC AUC (95% CI)
EMBEDDED	Logistic regression	0.657 (0.643–0.672)
EMBEDDED	Random forest	0.623 (0.608–0.639)
EMBEDDED	Gradient boosting	0.632 (0.618–0.647)
EMBEDDED	XGBoost	0.636 (0.621–0.651)
FEATURE	Logistic regression	0.629 (0.613–0.644)
FEATURE	Random forest	0.644 (0.630–0.659)
FEATURE	Gradient boosting	0.645 (0.629–0.660)
FEATURE	XGBoost	0.649 (0.634–0.664)

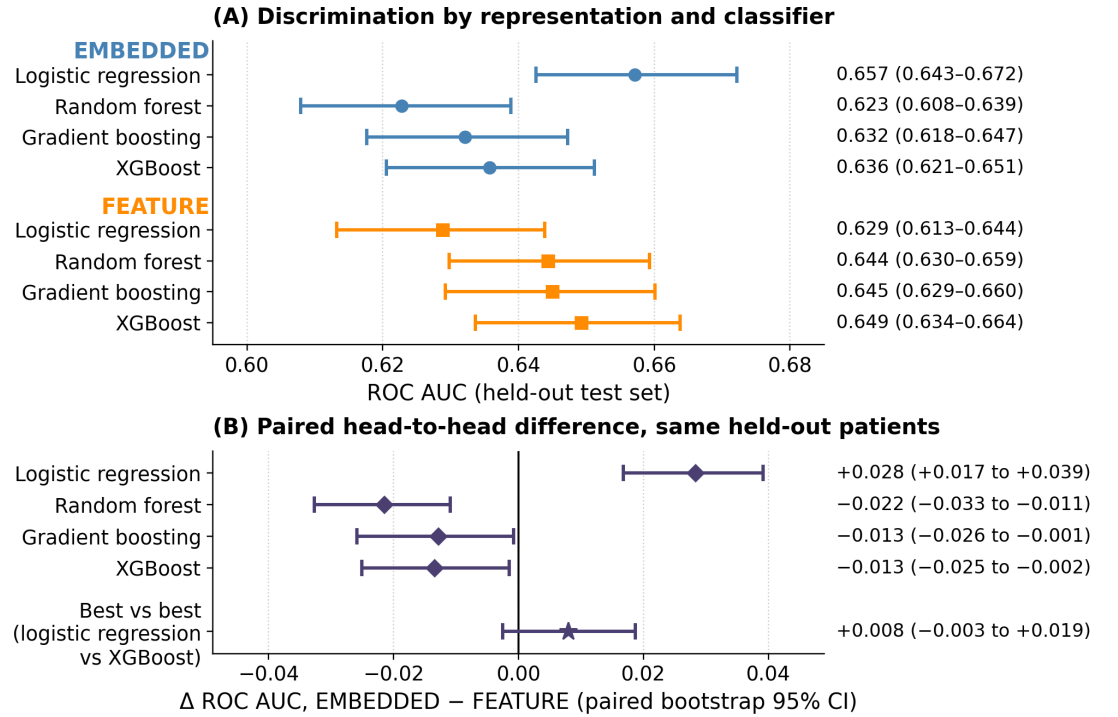
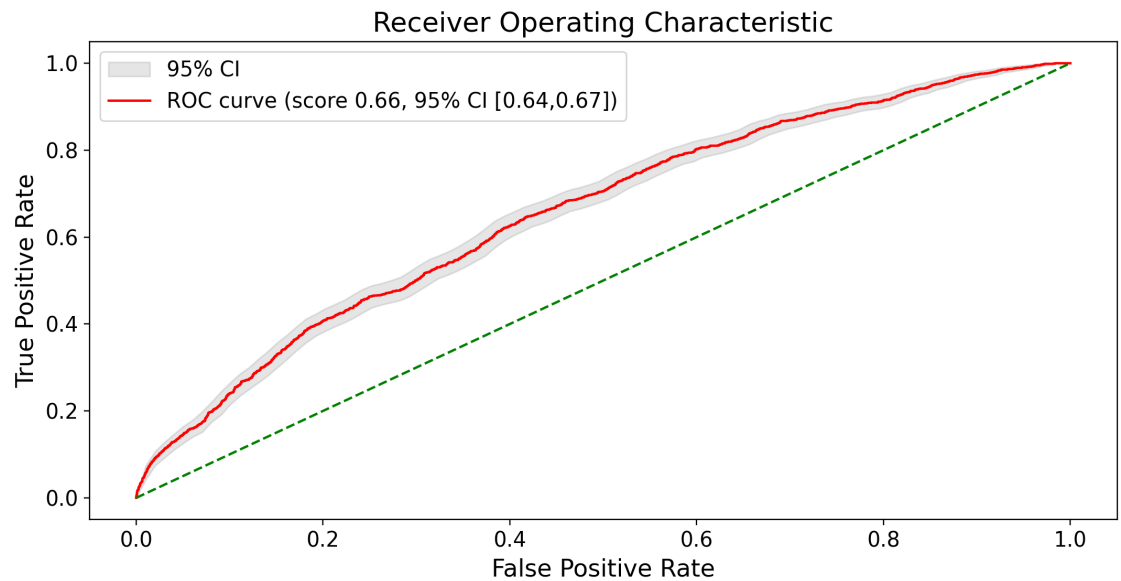


Figure 2. Discrimination by representation and classifier (held-out test set, $n = 8,516$; primary Qwen3-Embedding-8B encoder). **(A)** All eight representation-by-classifier combinations, grouped with EMBEDDED above FEATURE and in the same classifier order within each group, so the two can be read row for row. Markers are ROC AUC point estimates, bars are bootstrap percentile 95% confidence intervals, and the values are in Table 2. **(B)** The paired difference in ROC AUC, EMBEDDED minus FEATURE, with a paired bootstrap 95% interval. The solid line marks zero. The four diamonds hold the classifier fixed and vary only the representation; the star is the post-hoc best-feature-versus-best-embedded contrast, embedded logistic re-

gression minus feature-vector XGBoost.

(A) Embedded logistic regression



(B) Feature-vector XGBoost

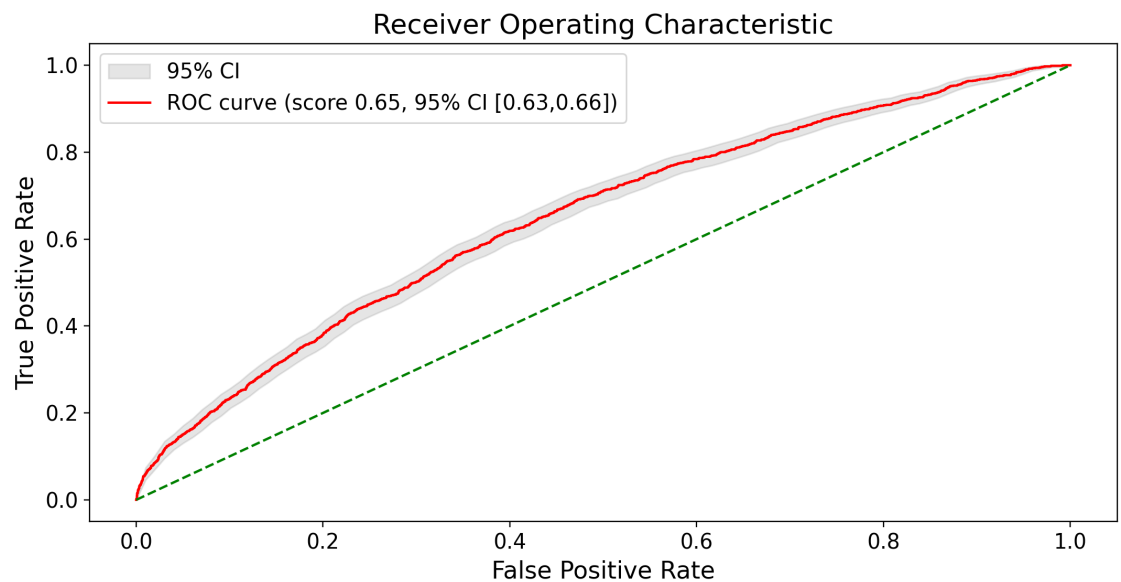
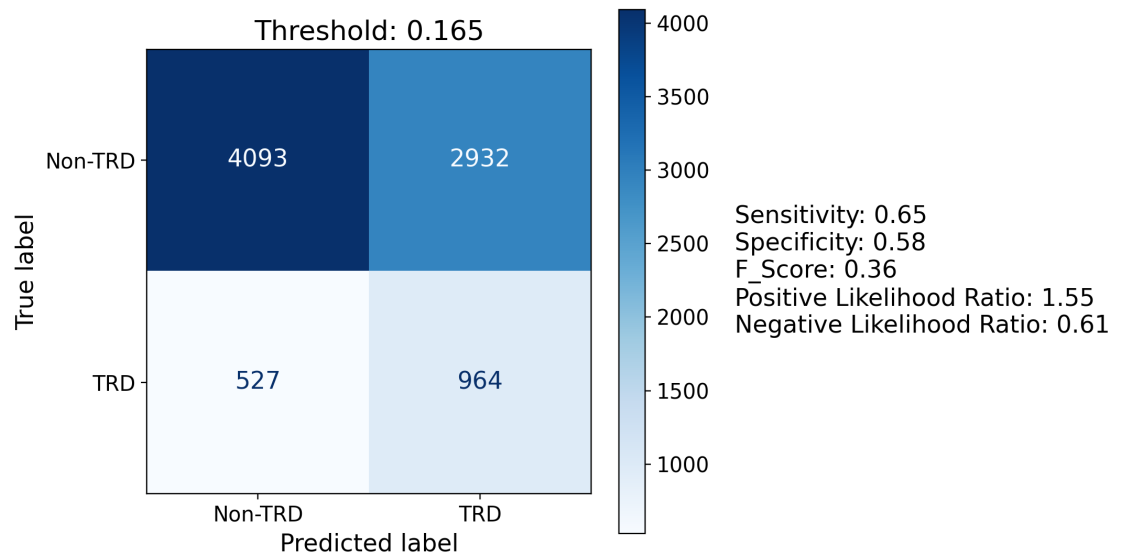


Figure 3. Receiver-operating-characteristic curves for the best

classifier on each representation (held-out test set; primary Qwen3-Embedding-8B encoder). Shaded bands are bootstrap 95% confidence intervals; point AUCs are given in Table 2.

The two behaved alike at their Youden-J operating points (Figure 4). Embedded logistic regression ran at slightly higher sensitivity than feature-vector XGBoost (0.65 versus 0.62), flagging 964 of 1,491 held-out TRD patients against 917, and at slightly lower specificity (0.58 versus 0.61), producing more false positives (2,932 versus 2,768). Both thresholds fell near 0.17. They were chosen on the same patients the models were scored on, so these figures describe the shape of each ROC curve rather than performance at a prespecified threshold, and are not candidate operating characteristics.

(A) Embedded logistic regression



(B) Feature-vector XGBoost

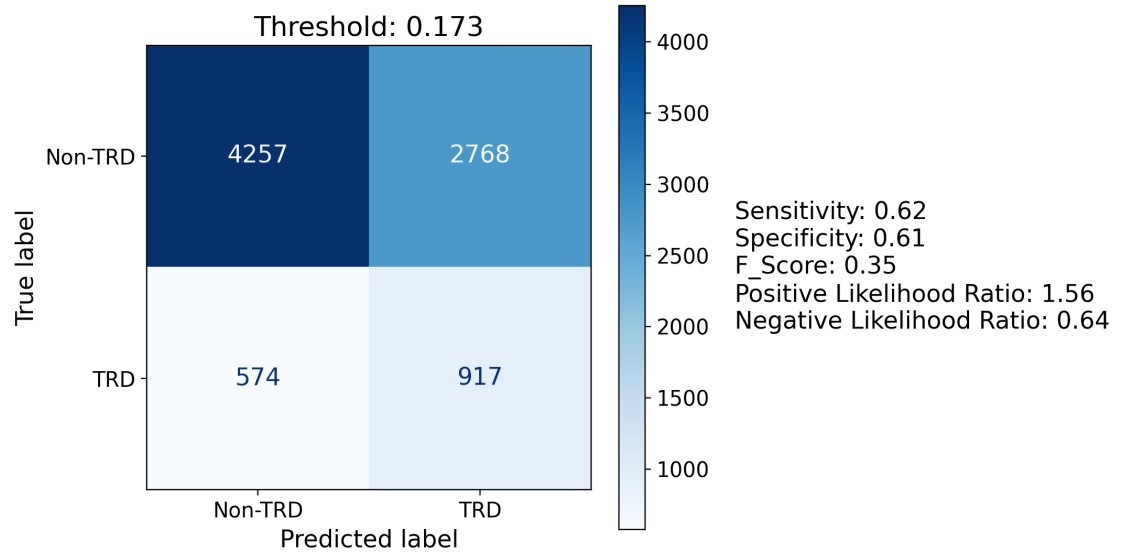


Figure 4. Confusion matrices at the Youden-J optimal operating point for the best classifier on each representation (held-out test set; primary Qwen3-Embedding-8B encoder).

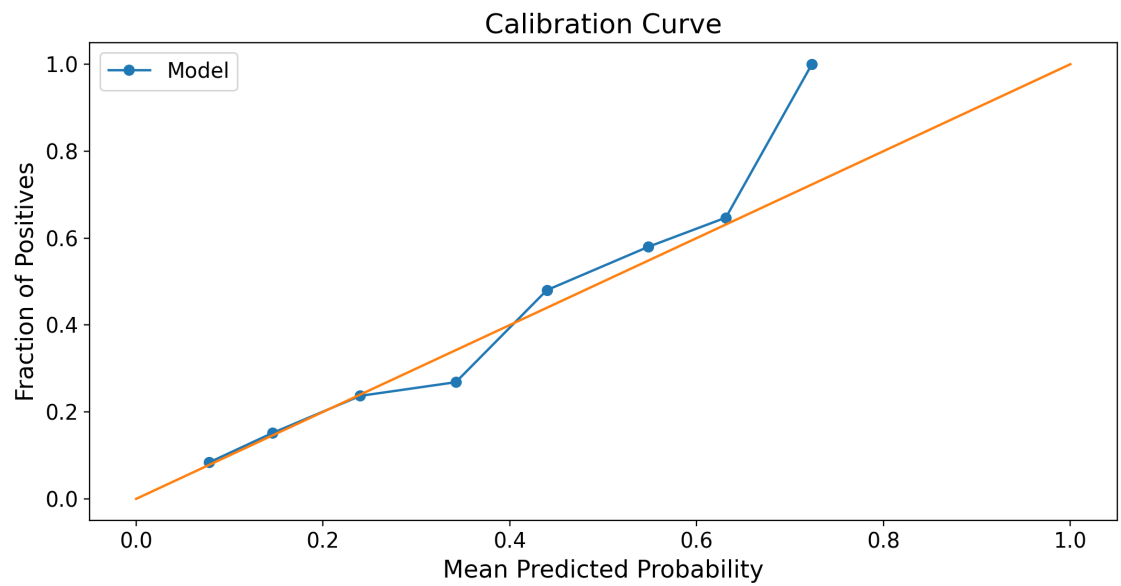
Model calibration

Neither representation was systematically better scaled than the other (Table 3, Figure 5). On the embedded representation, gradient boosting was the best-calibrated model (slope 1.01, intercept 0.01). Logistic regression was mildly under-confident (slope 1.27, the slope above 1 indicating probabilities pulled toward the base rate), random forest similar (1.15), and XGBoost mildly over-confident (slope 0.75). On the feature vector, XGBoost was best-calibrated (slope 1.02, intercept 0.02) and random forest the worst (slope 1.84). Each representation therefore contained both a near-ideally calibrated model and a poorly calibrated one, and which classifier occupied which position differed between them. No model placed any test prediction above 0.9, as expected at this base rate. Absolute-error metrics are in Supplement S3. Every one of these figures is scaled to this cohort's 17.5% TRD rate, which the sampling design chose, so none of them establishes calibration anywhere else.

Table 3. Calibration slope (ideal 1) and intercept (ideal 0) of the four classifiers on each representation (held-out test set). Brier score and weighted calibration error are reported in Supplement S3 (Table S3).

Representation	Classifier	Slope	Intercept
EMBEDDED	Logistic regression	1.27	−0.07
EMBEDDED	Random forest	1.15	−0.02
EMBEDDED	Gradient boosting	1.01	0.01
EMBEDDED	XGBoost	0.75	0.07
FEATURE	Logistic regression	0.78	0.06
FEATURE	Random forest	1.84	−0.15
FEATURE	Gradient boosting	0.87	0.04
FEATURE	XGBoost	1.02	0.02

(A) Embedded logistic regression



(B) Feature-vector XGBoost

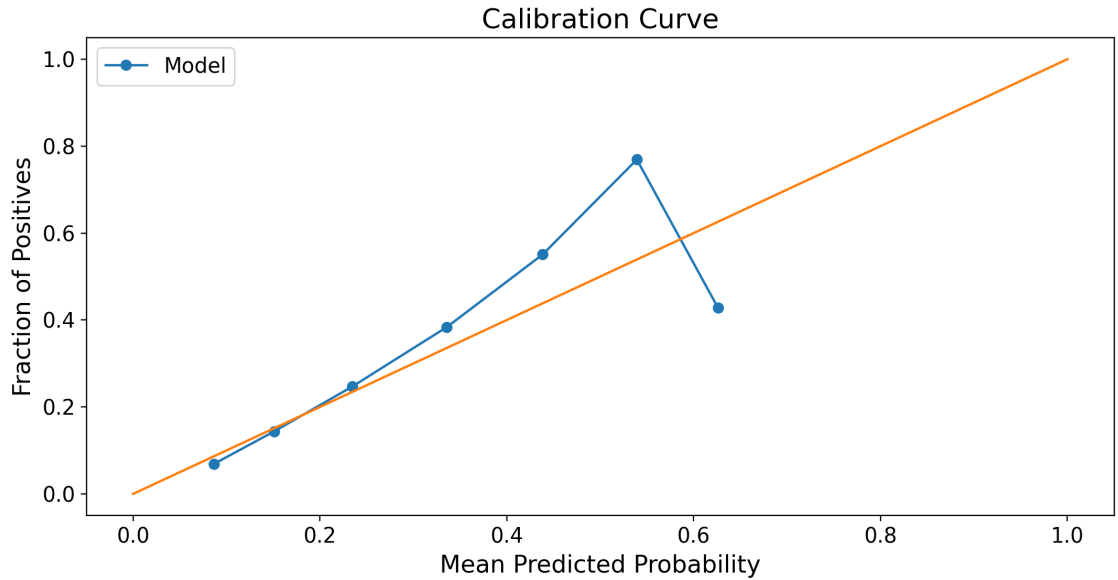
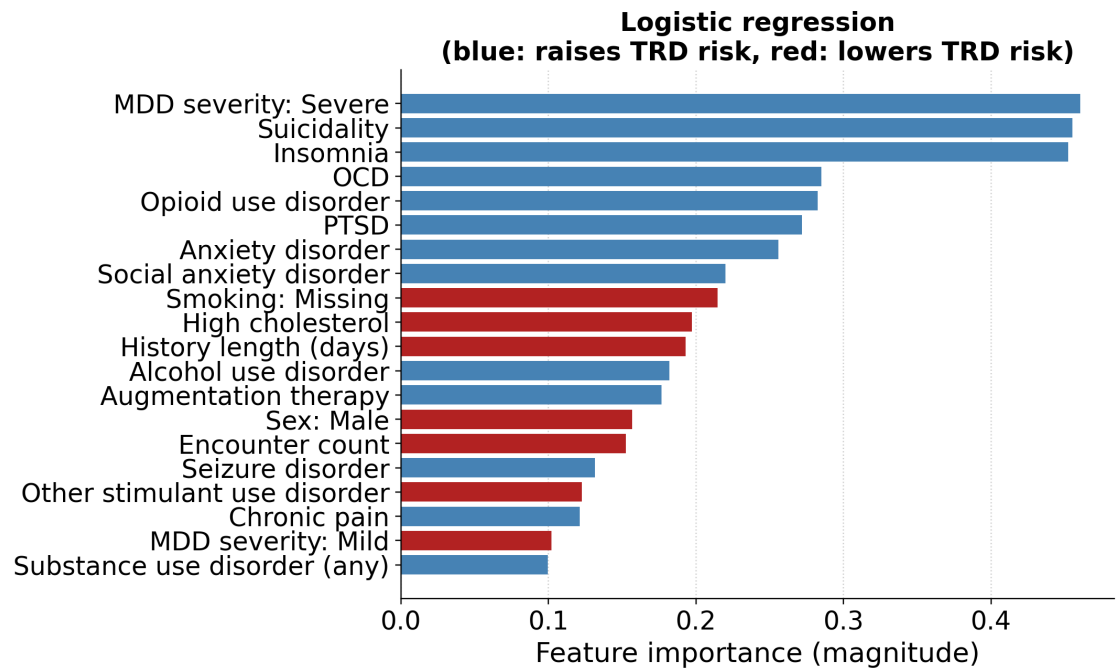


Figure 5. Calibration curves for the best classifier on each representation (held-out test set; primary Qwen3-Embedding-8B encoder). The diagonal is perfect calibration. Deviations above it indicate under-confidence and below it over-confidence. Per-classifier calibration slope and intercept are given in Table 3.

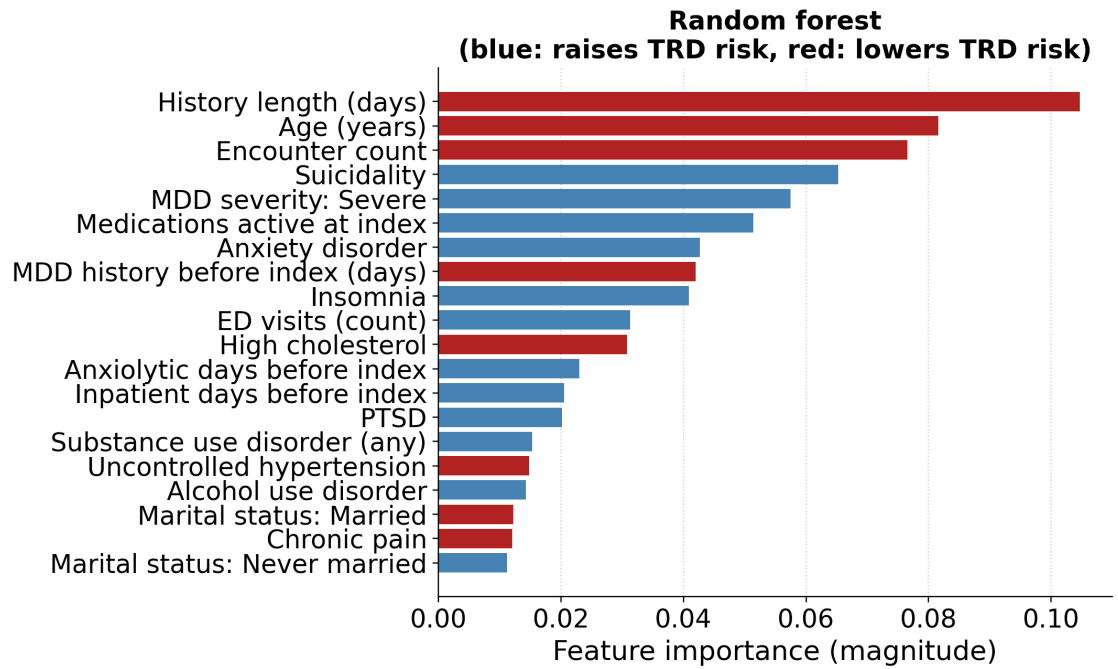
What each representation reads

Only the feature vector can be read directly. The embedded representation’s 4,096 latent dimensions carry no individual clinical meaning, so per-concept attribution on the embedded side is addressed by the ablation below rather than by an importance ranking. On the feature vector, the highest-weighted predictors recapitulated the strongest univariate correlates of TRD in Table 1. Severe MDD coding, a flagged suicidality history, insomnia, obsessive-compulsive disorder, opioid use disorder, PTSD, and anxiety disorder carried the largest positive logistic-regression weights. Indicators such as missing smoking status, hyperlipidemia, longer pre-index history, and male sex carried negative weights. Importances for all four classifiers are shown in Figure 6. The tree ensembles concentrated on a similar set of psychiatric and substance-use predictors at the top, with direction of effect recovered by univariate correlation. The readable side of the null is therefore reading recognizable psychiatric risk, which is what separates a tie between two working models from a tie between two failing ones.

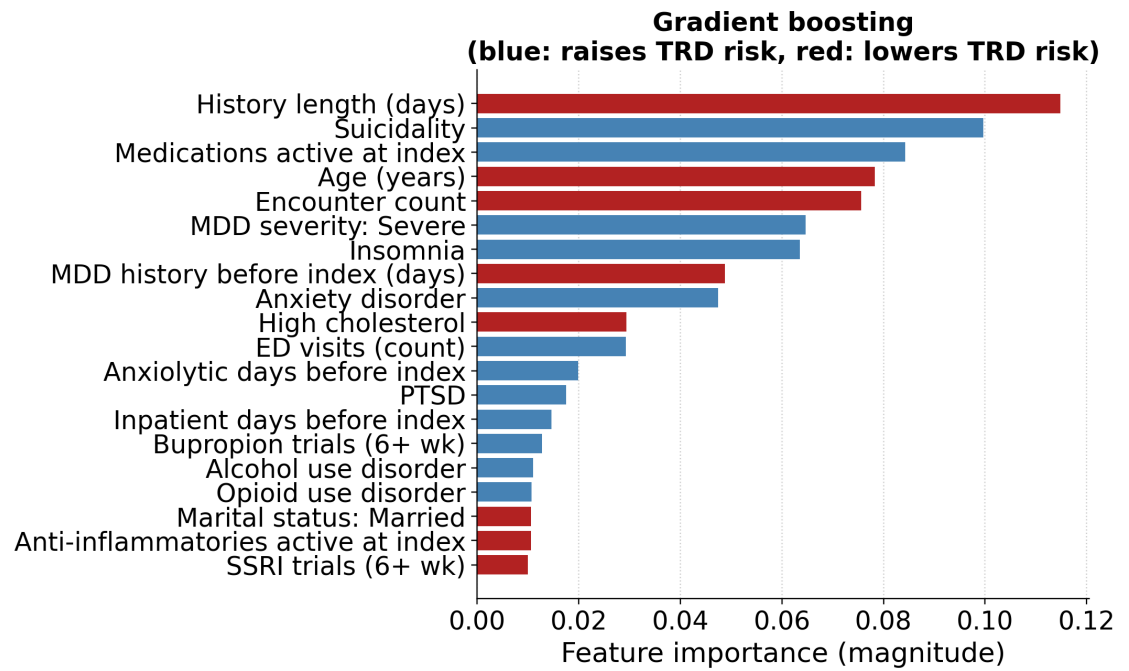
(A) Logistic regression



(B) Random forest



(C) Gradient boosting



(D) XGBoost

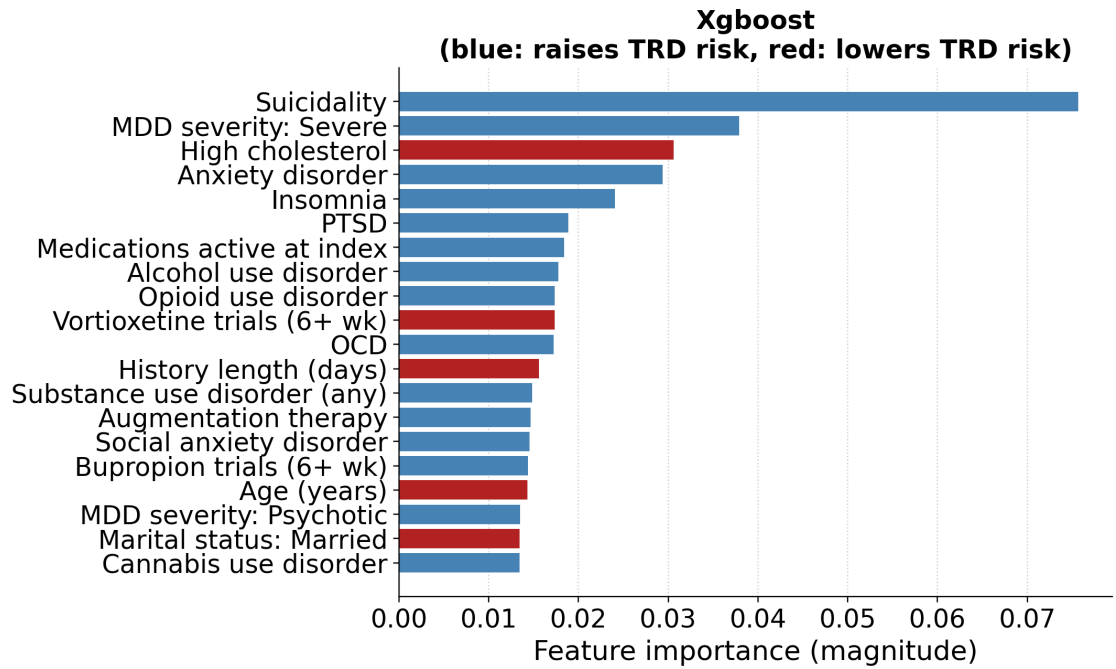


Figure 6. Feature importance on the feature-vector representation, one panel per classifier. (A) Signed logistic-regression coefficients (steelblue raises TRD risk, firebrick lowers it). (B) Random forest, (C) gradient boosting, and (D) XGBoost importances, with direction-of-effect recovered by univariate correlation. The highest-ranked predictors mirror the largest TRD-stratified differences in Table 1.

The embedded model used far fewer dimensions than it was given. Cross-validated elasticnet left 385 of the 4,096 with nonzero weight, and roughly 179 of those carried 80% of the total coefficient magnitude (roughly 236 carried 90%; Figure 7). Fewer than 5% of the available dimensions therefore account for most of the fit, and model-agnostic correlation and principal-component checks agree (Supplement S1).

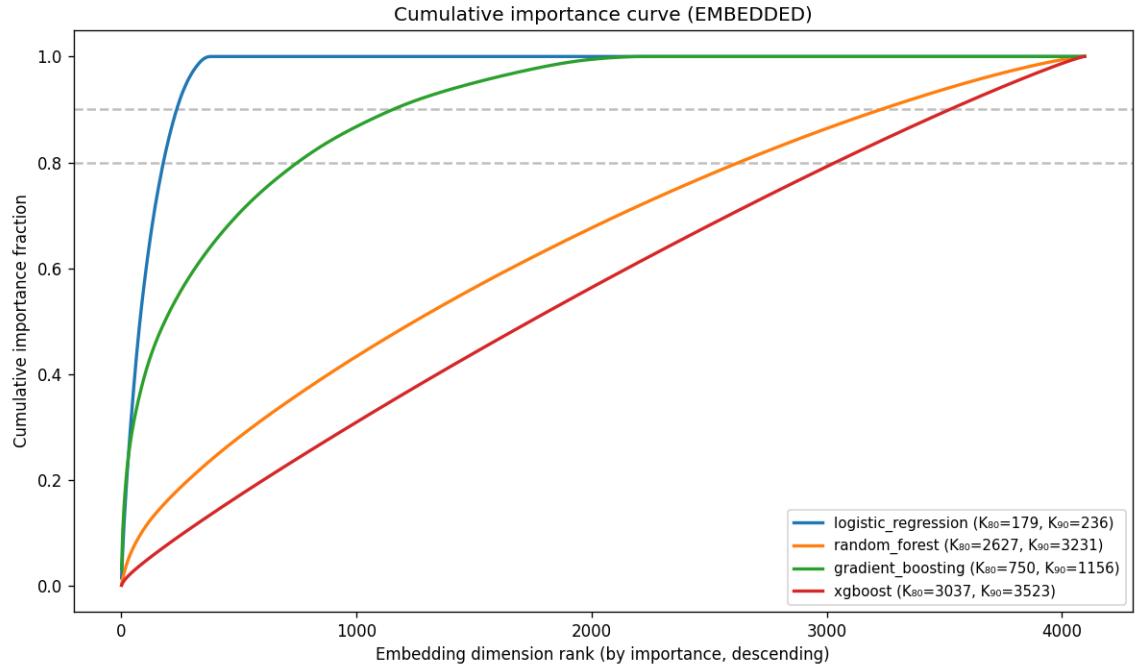


Figure 7. Effective dimensionality of the embedded representation (primary Qwen3-Embedding-8B encoder): cumulative built-in feature importance for the four embedded classifiers. Each curve plots the cumulative fraction of total importance mass against dimension rank, with dimensions sorted by descending native importance, and the K_{80} / K_{90} knees are reported in the legend. Model-agnostic correlation and PCA-based checks are in Supplement S1.

Semantic-feature ablation

Permuting narrative concepts one at a time located the embedded signal in psychiatric history, medication burden, and prior treatment exposure. Psychiatric history produced the largest discrimination loss (logistic regression, a ROC AUC change of -0.028 , with the other three classifiers between -0.024 and -0.027). Medication burden produced the second largest (-0.003 to -0.027) and treatment exposure the third (-0.019 for logistic regression, and $+0.000$ to -0.011 for the other three). Permuting the two sociodemographic concepts, race/ethnicity and social determinants of health, or the treatment-contraindication section changed discrimination by no more than 0.005 under any classifier (Table 4, Figure 8).

Paired-bootstrap intervals excluded zero for psychiatric history in all

four classifiers, medication burden in three, and treatment exposure in two. Among the remaining concepts only one interval excluded zero, treatment contraindications under XGBoost at -0.005 (-0.008 to -0.001), and race and social determinants moved ROC AUC by less than 0.003 anywhere.

The domains the embedding relies on are the same domains that carry the largest feature-vector weights. That convergence is the mechanism behind the null in Table 2: the two representations are reading the same clinical content out of the same record, and the embedding reorganizes it without adding to it.

Table 4. *Semantic-feature ablation: change in ROC AUC versus the frozen baseline when each narrative concept is permuted across donors (embedded representation, primary Qwen3-Embedding-8B encoder). More negative = larger reliance on that concept. Point estimates, ordered by descending logistic-regression loss to match Figure 8. Paired-bootstrap intervals are quoted in the text.*

Permuted concept	LR	RF	GB	XGB
Psychiatric history	-0.028	-0.027	-0.024	-0.027
Medication burden	-0.027	-0.003	-0.014	-0.017
Treatment exposure	-0.019	$+0.000$	-0.006	-0.011
Treatment contraindications	-0.002	$+0.001$	-0.001	-0.005
Social determinants (SDOH)	-0.000	$+0.000$	$+0.000$	$+0.001$
Race/ethnicity	-0.000	$+0.000$	-0.003	-0.000

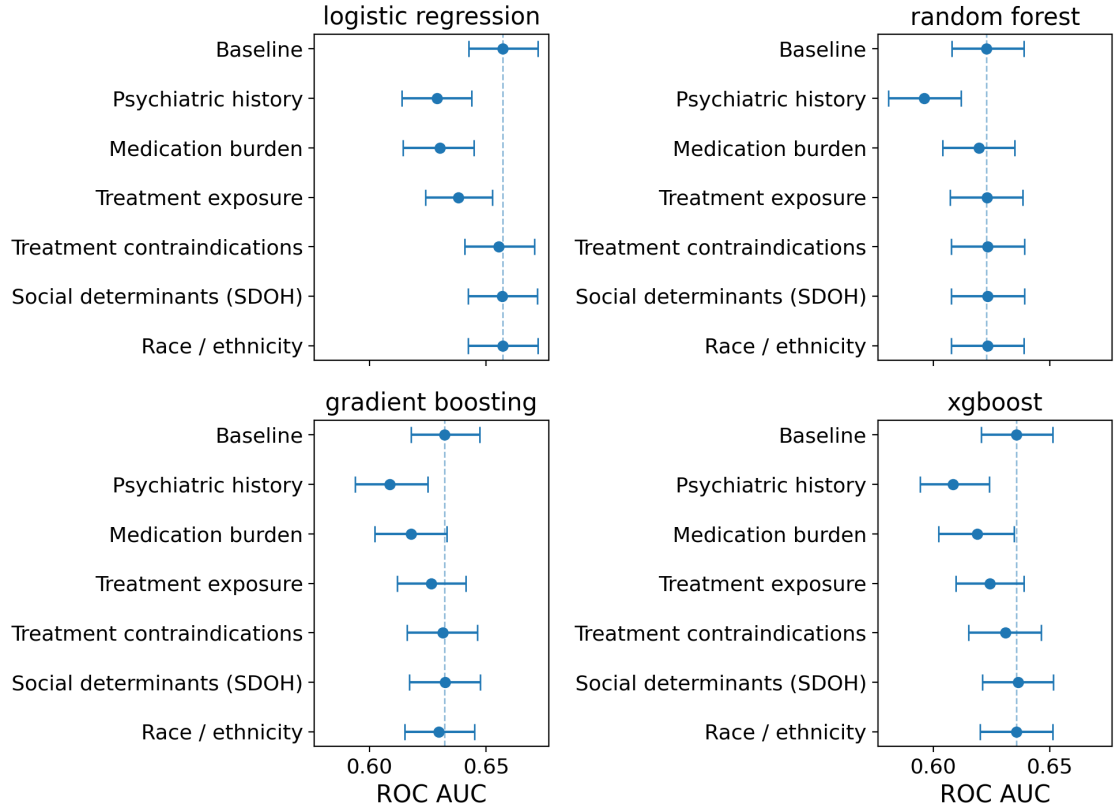


Figure 8. Semantic-feature ablation, absolute-discrimination view (embedded representation, primary Qwen3-Embedding-8B encoder). Each row is a run: the unablated baseline on top, then the six permutation specifications ordered by descending logistic-regression ROC AUC drop, the same ordering reused across panels. There is one panel per classifier, arranged two by two on a shared ROC AUC axis, with a reference line at the baseline ROC AUC. Permuting psychiatric history, medication burden and treatment exposure produces the largest discrimination loss, whereas the remaining three concepts move ROC AUC comparatively little (deltas in Table 4).

Robustness across encoders

The comparison did not depend on which encoder produced the embedding. Across the four encoders the embedded logistic-regression model occupied a 0.012-wide discrimination band (ROC AUC 0.645–0.657, Table 5 and Figure 9A). Qwen3-Embedding-8B was highest

(0.657, 95% CI 0.643-0.672), with bge-en-icl and Qwen3-Embedding-4B tied just behind (both 0.655) and bge-small-en-v1.5 lowest (0.645). Logistic regression was the best-discriminating classifier on the embedded representation for every encoder. Only bge-small-en-v1.5 scored below the best feature-vector model. The other three scored just above it, and all four fall inside that same band, which is narrower than every confidence interval in Table 2. The choice of encoder therefore does not change what Table 2 shows.

The ablation also reproduced across encoders (Figure 9B). Psychiatric history and medication burden cost the most on all four, -0.027 to -0.030 and -0.022 to -0.034 respectively. Every one of those intervals excluded zero. The two sociodemographic permutations moved ROC AUC little. The order of the two largest concepts held everywhere except in bge-en-icl, where medication burden cost more than psychiatric history, and it did so for all four classifiers there (embedded logistic regression -0.034 versus -0.027).

Table 5. *Embedded logistic-regression discrimination by encoder (held-out test set). 95% CIs are bootstrap percentile intervals. “Dimensions” is the encoder’s output dimensionality, which is also the number of predictor columns the classifier receives; EPV is events per variable, 5,964 TRD-positive training patients divided by that column count (Supplement M7). Note that EPV runs opposite to discrimination: the encoder with the most favorable EPV is the least discriminating.*

Encoder	Dimensions	EPV	ROC AUC (95% CI)
bge-small-en-v1.5	384	15.5	0.645 (0.629-0.660)
bge-en-icl	4,096	1.5	0.655 (0.641-0.670)
Qwen3-Embedding-4B	2,560	2.3	0.655 (0.641-0.670)
Qwen3-Embedding-8B	4,096	1.5	0.657 (0.643-0.672)

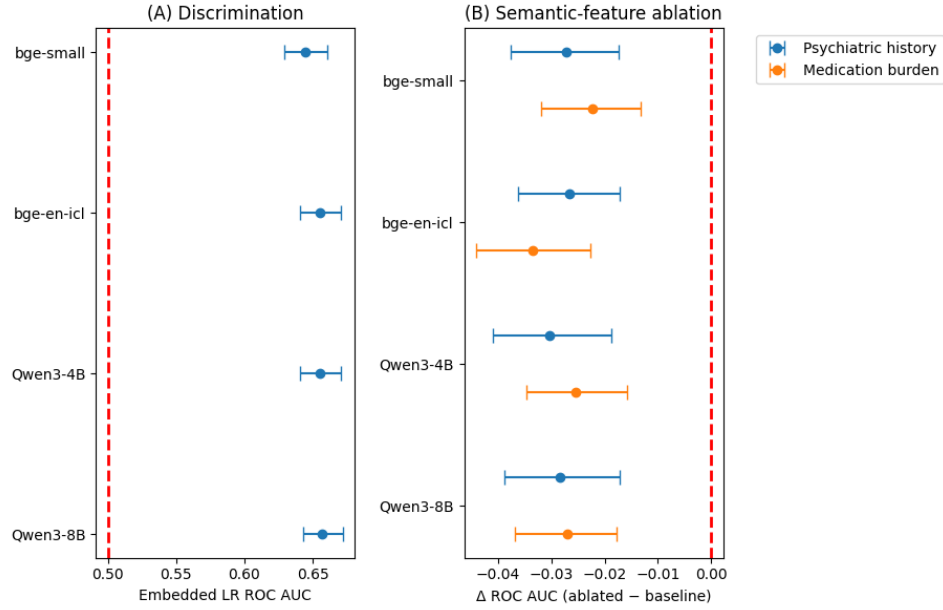


Figure 9. Cross-embedder robustness: embedded logistic-regression ROC AUC (A) and the two largest semantic-feature ablation deltas (psychiatric history, medication burden; B) across all four encoders (bge-small-en-v1.5, bge-en-icl, Qwen3-Embedding-4B, Qwen3-Embedding-8B), demonstrating that the principal conclusions hold beyond the Qwen3-Embedding-8B encoder. Error bars are bootstrap (A) and paired-bootstrap (B) 95% confidence intervals.

Prediction from retrieved neighbors

Retrieval recovered real signal from the embedding geometry. Retrieving the nearest neighbors discriminated at ROC AUC 0.594 (95% CI 0.578-0.610) under cosine weighting. Random retrieval landed at chance (0.499, 95% CI 0.483-0.515). A space in which a patient’s closest analogues predict the outcome and randomly drawn patients do not is a space that carries label information.

How neighbors were weighted mattered far less than which neighbors were retrieved. Uniform and cosine weighting differed by at most 0.005 ROC AUC within either retrieval scheme, against the gap from 0.594 to 0.499 between nearest and random retrieval (Table 6). Once the 50 retrieved patients are already the most similar available, adjusting their relative contributions has almost nothing left to do.

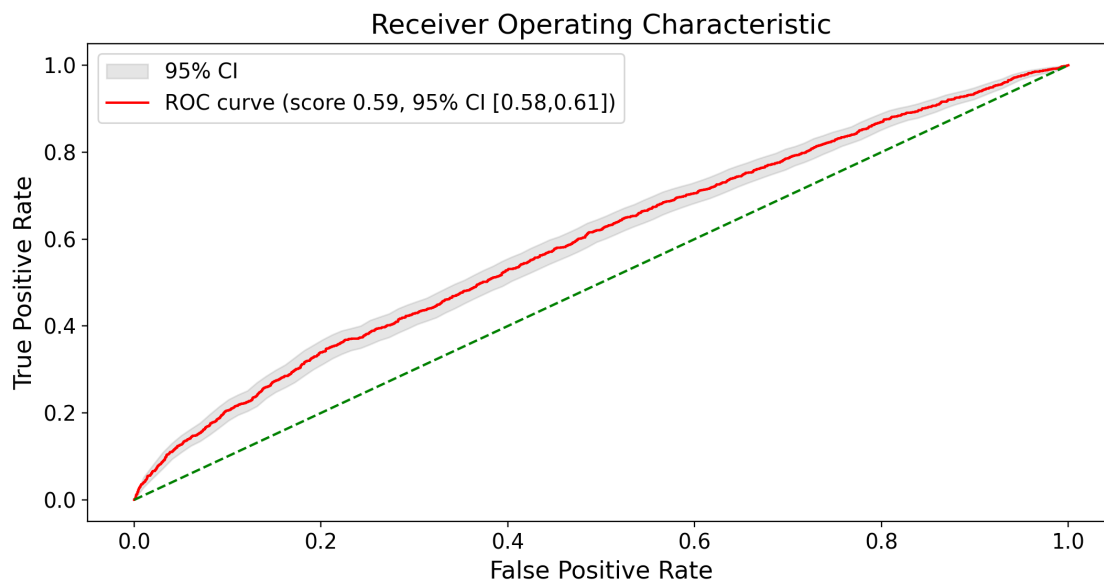
Retrieval nevertheless lost decisively to a model fitted on the same

embedding. The best retrieval configuration reached 0.594 (0.578–0.610) against 0.657 (0.643–0.672) for embedded logistic regression, a shortfall of 0.063 ROC AUC on intervals that do not overlap. The comparison is unpaired and therefore conservative, since both predictors score the same held-out patients. Retrieval also fell below every one of the eight trained configurations in Table 2, including the weakest of them at 0.623.

Table 6. Neighbor-weighted ROC AUC by retrieval scheme and weighting strategy (embedded representation, primary Qwen3-Embedding-8B encoder, held-out test set; $K = 50$ neighbors). 95% CIs are bootstrap percentile intervals. Random retrieval is the negative control. For comparison, embedded logistic regression on the same representation and the same patients reaches 0.657 (0.643–0.672).

Retrieval scheme	Uniform (95% CI)	Cosine (95% CI)
Nearest	0.593 (0.578–0.609)	0.594 (0.578–0.610)
Random	0.495 (0.479–0.510)	0.499 (0.483–0.515)

(A) Nearest retrieval



(B) Random retrieval

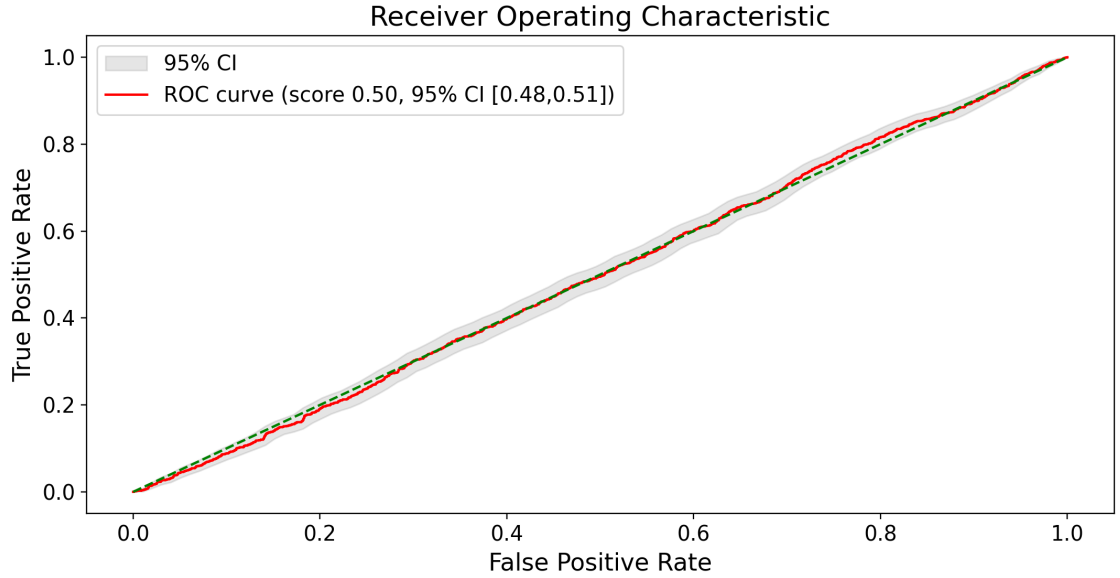


Figure 10. Neighbor-weighted ROC for the cosine-weighted predictor on the embedded representation (held-out test set, $n = 8,516$, $K = 50$; primary Qwen3-Embedding-8B encoder). (A) Nearest retrieval; (B) random retrieval. The separation between the two panels is the discriminative signal in the embedding geometry. Shaded bands are bootstrap 95% confidence intervals and the dashed diagonal is chance. The uniform weighting is in Table 6.

Both halves of that result reproduced on every encoder (Table 7). Nearest retrieval beat random retrieval by 0.083 to 0.095 ROC AUC on all four, and on all four it fell short of that encoder’s own trained logistic regression by 0.061 to 0.065. The gap between retrieval and a fitted model is therefore a property of the approach rather than of the primary encoder.

Table 7. Neighbor-weighted retrieval against embedded logistic regression, by encoder (cosine weighting, $K = 50$, held-out test set). 95% CIs are bootstrap percentile intervals. Random retrieval is the negative control. The last column is that encoder’s own model from Table 5, scored on the same patients.

Encoder	Nearest (95% CI)	Random (95% CI)	Embedded logistic regression (95% CI)
bge-small-en-v1.5	0.580 (0.564–0.596)	0.497 (0.482–0.513)	0.645 (0.629–0.660)
bge-en-icl	0.592 (0.577–0.607)	0.498 (0.482–0.515)	0.655 (0.641–0.670)
Qwen3-Embedding-4B	0.594 (0.579–0.611)	0.500 (0.484–0.516)	0.655 (0.641–0.670)
Qwen3-Embedding-8B	0.594 (0.578–0.610)	0.499 (0.483–0.515)	0.657 (0.643–0.672)

Validity checks

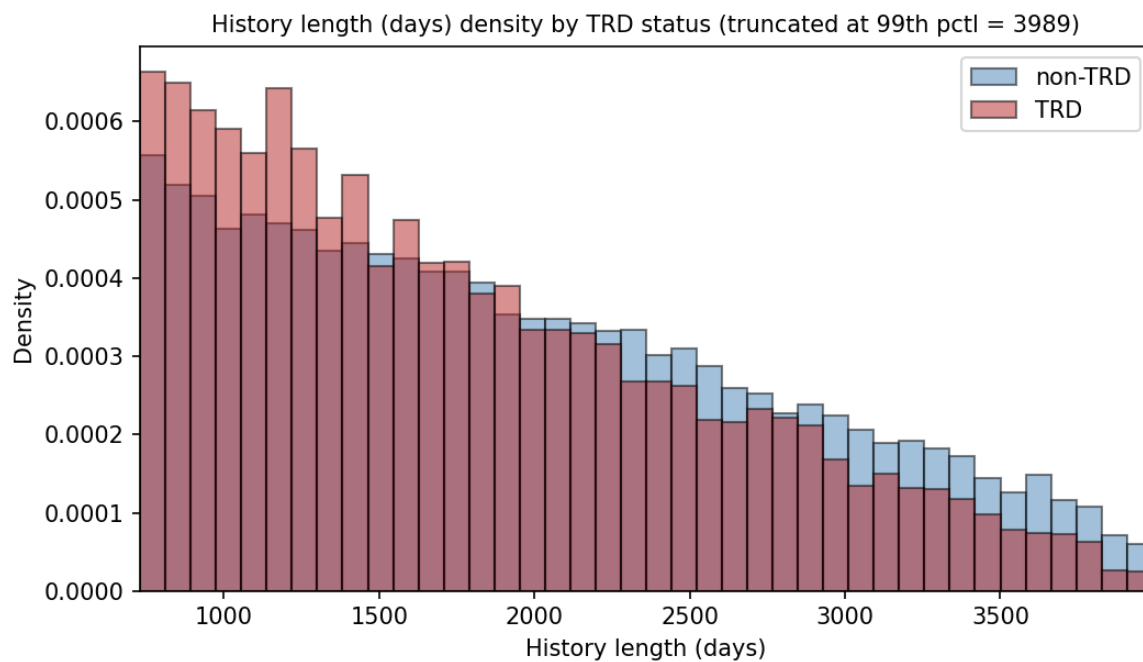
Three checks bear on both objectives, because an artifact in any of them would undermine the comparison as much as the retrieval result. None of the three found one.

Train/test comparability. The training and test sets were closely matched on every predictor. Over 100 predictor rows the largest absolute standardized mean difference was 0.036, none reached the conventional 0.1 threshold, and the maximum sat on a social-determinant flag recorded for 22 training patients and no held-out patient. That is a small-cell artifact rather than a distributional difference. No measured difference separates the evaluation sample from the development sample, and neither is representative of a wider population, because both halves inherit the extract’s case enrichment. Supplement S6 tabulates the distributions in the form of Table 1.

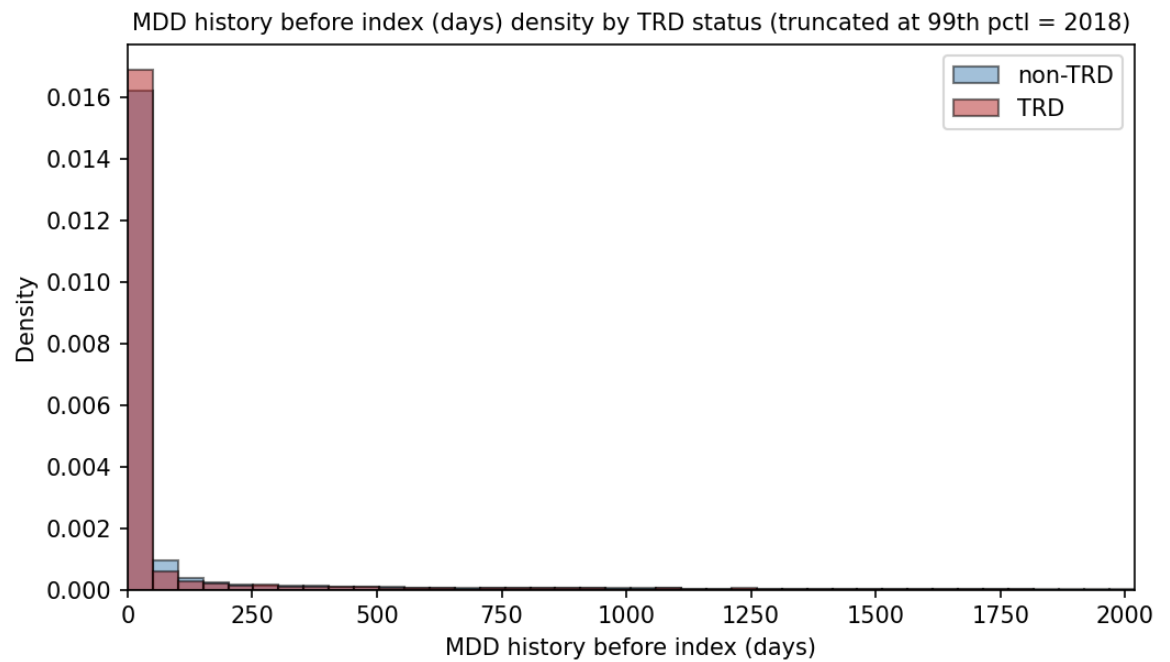
Data volume did not act as a confound. Patients with richer records could appear higher-risk for reasons that have nothing to do with depression, so we correlated the TRD label with three volume and recency proxies on the full untruncated data. All three associations were weak and negative: pre-index history length (Spearman $\rho = -0.073$), encounter count ($\rho = -0.064$), and MDD-to-index gap ($\rho = -0.029$). TRD-positive patients did not simply have more data (Figure 11A–C), and discrimination is flat across quintiles of pre-index history length, which is the stronger form of the same test (Supplement S5). Most patients were prescribed an antidepressant on the day of their MDD diagnosis ($n = 27,906$). Their TRD rate was 18.4%

against 15.9% for the 14,673 with a delayed prescription, both near the 17.5% base rate (Figure 11D).

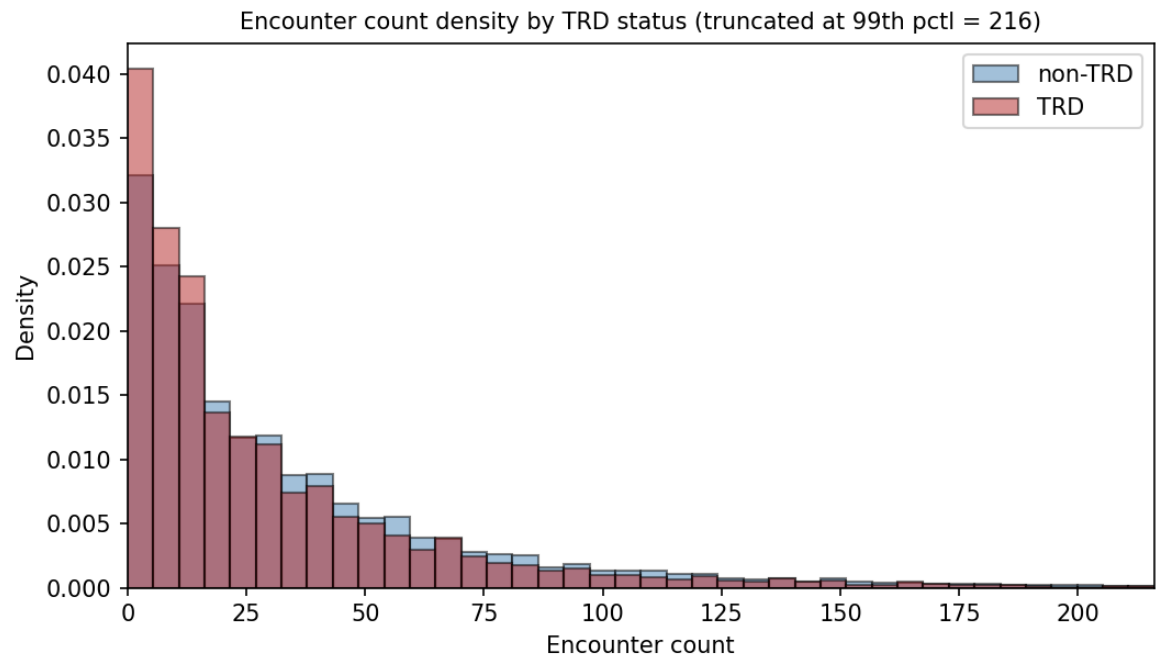
(A) Pre-index history length



(B) MDD-to-index gap



(C) Encounter count



(D) TRD rate, same-day vs delayed prescription

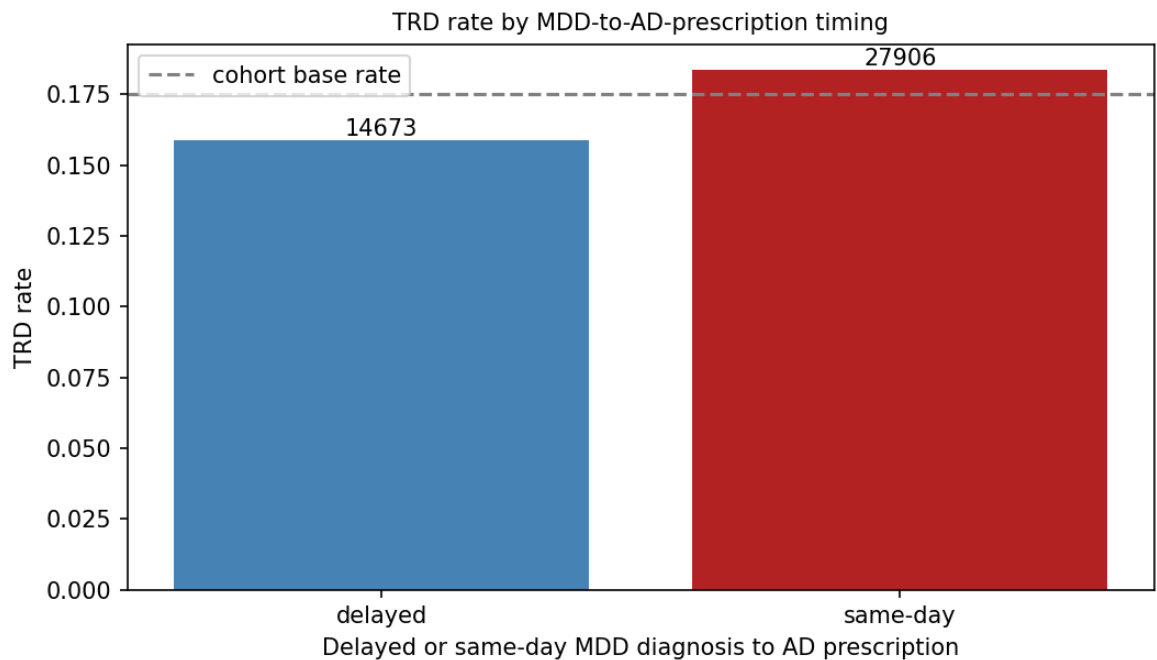


Figure 11. Volume and recency confound checks on the full untruncated data. (A-C) TRD-stratified distributions of the three proxies: (A) pre-index history length; (B) MDD-to-index gap; (C) encounter count. (D) TRD rate among patients with a same-day versus delayed MDD-to-antidepressant prescription (dashed line = 17.5% cohort base rate; bar labels are group n).

Subgroup performance. Most of what separates subgroups here is how the depression is coded, not who the patient is. The same models were scored again inside eight strata, with nothing refit, giving 240 between-group contrasts across the three arms. At a conventional 5% threshold about twelve of those clear the bar by chance alone, so P values were adjusted across the whole set by Benjamini-Hochberg at a 5% false discovery rate. Fifty-eight contrasts exclude zero before that adjustment and 24 survive it, 21 of them about MDD coding. Discrimination is higher where MDD is coded recurrent or severe (+0.059 to +0.087 ROC AUC) and lower where it is coded single-episode (−0.074 to −0.056), in all three arms. That is a documentation effect as much as a phenotype one: the models do better where the diagnosis is recorded specifically, and worse where it is left unspecified, which is most of this cohort.

The sociodemographic strata gave three surviving contrasts, and all three are consistent with thinner records supporting weaker prediction. Discrimination is lower for patients aged 18–29 in the feature-

vector arm (-0.068 , adjusted $P = .04$). It is also lower for never-married patients in the embedded arm, where two of the four classifiers survive, at -0.054 and -0.051 (adjusted $P = .01$). No sex contrast came close: all ten include zero and none exceeds 0.012 ROC AUC. Preferred language was not analyzed, because 98.9% of the cohort prefers English.

The data neither establish nor rule out a race-associated performance gap. All ten White-minus-non-White contrasts are positive ($+0.005$ to $+0.053$) and five exclude zero before adjustment, but none survives it (smallest adjusted $P = .11$). The minority stratum carries 302 outcome events against 1,181, which leaves its intervals roughly twice as wide. Feature-vector calibration is worse among non-White patients (slope 0.79 versus 0.98) while the embedded arm holds (0.95 versus 0.97). Supplement S7 has the full stratified results, and Limitations takes up what a gap this wide can and cannot be read to mean.

Discussion

Principal findings

The embedding did not beat the feature vector. Embedded logistic regression reached 0.657 against 0.649 for the best feature-vector model. Both scored the same held-out patients, so the paired comparison is the one that counts: $+0.008$, on an interval from -0.003 to $+0.019$. Neither representation is better than the other. Both were built from the same hand-picked record, which Supplement S8 lists field by field, so this says nothing about whether narrative adds clinical information. It says an embedding can reorganize the same signal without losing discrimination, and it says nothing about what an encoder would find in a raw record.

Three results point the same way. The ablation traces the embedded model's reliance to psychiatric history, medication burden and prior treatment exposure, which are the domains carrying the largest feature-vector weights. The embedded model uses 385 of 4,096 dimensions, with 80% of the coefficient magnitude in roughly 179 of them. And discrimination stays within 0.012 across four encoders.

It is an interaction, not a tie. The embedding beat the feature vector with logistic regression ($+0.028$) and lost to all three tree ensembles (-0.013 to -0.022). Dense coordinates suit a regularized linear model. Trees split one coordinate at a time, and those coordinates mean nothing on their own. A study comparing representations

under a single learner could report either result, depending on which learner it picked.

Retrieval works, and loses. Nearest retrieval discriminated far better than random on every encoder, so proximity in the embedding tracks TRD risk. It still fell 0.063 short of a model fitted to the same embedding, on intervals that do not overlap, and below all eight trained configurations. Averaging the outcomes of a patient's 50 closest analogues is a worse use of the representation than fitting a model to it.

Two things cause that gap. A fitted model can ignore the dimensions carrying no outcome signal, which is what selecting 385 of 4,096 does, whereas cosine similarity weights all 4,096 equally. And 50 neighbors is a noisy way to estimate a rate near 0.175, however well they are chosen. The first points toward a similarity learned against the outcome. The second is a variance cost retrieval always pays.

Clinical interpretation

The strongest feature-vector predictors are the ones a clinician would expect: a history of suicidality, severe MDD coding, and broad psychiatric and substance-use comorbidity. The ablation lands on the same domains. That is reassuring for face validity and nothing more, because every one of those predictors can equally reflect how much treatment a patient was able to get. That matters because the outcome is a treatment trajectory, not a verified failure to respond. Definitions of EHR-defined TRD differ in what they count, and the choice changes both how common it looks and who ends up in the group [6-8].

The temporal design is a real strength. Every predictor was fixed before the index prescription and the treatment sequence defining the outcome came afterward. That is what stops a model from reconstructing the outcome rule out of data recorded alongside it, which is the mechanism suspected behind the sharp external-validation drops reported elsewhere [12].

The sociodemographic ablation needs the same reading. It shows the models did not need the recorded race/ethnicity and social-determinant fields to reach their ROC AUC. It does not show the prediction pathway is demographically neutral. Utilization and treatment variables can carry differential access, and the chance to receive several medication changes is itself socially patterned.

Comparison with prior work

Reported discrimination for treatment resistance varies widely, and 0.65 is not a ceiling [18-21,30]. What moves it is the phenotype definition, the prediction horizon, how close the predictors sit to the treatment trajectory, and whether validation is internal or external. Liberman and colleagues reached 0.83 over 24 months in 35,246 adults, on predictors that mix illness burden with the intensity of care received [9]. Lage and colleagues reported 0.652 (95% CI 0.623-0.682) under external validation, with a top-quintile lift of 1.99 [10]. A single-site multimodal study reported 0.684 from structured data, 0.569 from notes and 0.728 from the two combined, with MRI highest [11].

Transportability is the harder constraint. Across three health systems, internal ROC AUCs of 0.58-0.64 fell to 0.51-0.58 under external validation, and patient-level risk estimates agreed poorly between sites [12]. The 0.65 reported here sits with the externally validated and single-site structural estimates. It is neither a new ceiling nor an improvement on any of them [9-12]. What this study contributes is the controlled comparison: the same patients, cutoff, source record, outcome and held-out split. Within it, the embedding preserved the signal a transparent feature vector already carried, and did not exceed it.

Limitations

The outcome counts switches, and cannot see why a switch happened. This is the binding limitation on everything above. The label is assigned when a patient receives three or more distinct antidepressant treatments within a year [2], and nothing in the record says whether the change followed a genuine failure to respond. A patient is switched because the drug did not work, or because it caused intolerable side effects, or because it was unaffordable, or because they stopped taking it, or because their prescriber prefers switching to augmenting. All of these increment the count identically, and this study cannot distinguish them. Symptom response was not systematically recorded, and dose and duration adequacy were not verified for the counted treatments. The target therefore approximates rather than establishes the consensus characterization of TRD as inadequate response to at least two adequate trials [4,5].

Switching also depends on continuity of care, insurance, and access. Those differ across demographic groups in the delivery of depression treatment in the United States [16,17], so the label can encode health-care process alongside pharmacologic non-response. Changing the

operational rules materially alters both prevalence and cohort composition [6-8].

The proxy is wrong in two directions. It counts patients who changed drugs for reasons that have nothing to do with whether the drug worked, such as a change in insurance or a break in care. It misses patients who stayed on one inadequate drug and never switched, and that is the group a clinician would worry about most. This study cannot see them at all. Both arms predict the same imperfect label, so the head-to-head comparison is not biased by it. What it costs is the meaning of the absolute number, which is discrimination against the proxy rather than against TRD. Two things would sharpen the target and neither was done here: checking the label against chart review, and building a stricter label that requires documented adequate courses after the index date.

Validation is internal, and the reported discrimination is an upper bound. This is a single random split of one health system’s records spanning 2016–2024. There is no temporal split and no external cohort, so nothing here speaks to transportability across time or site. The study also cannot detect the coding, documentation, and prescribing drift that accumulates over a nine-year window, because a random split distributes any such drift evenly across both sides and hides it. Reported discrimination should therefore be read as an upper bound on what the same models would achieve on later data or in another system. The drop seen elsewhere under external validation is substantial [12]. By construction the embedded representation also runs below the conventional events-per-variable threshold of 10 on three of the four encoders ($EPV \approx 1.5\text{--}2.3$), whereas the feature-vector model comfortably exceeds it at ≈ 65 .

Retrieval was tested in one form. The retrieval result bounds the digital-twin premise as implemented here, not the premise in general. Similarity was geometric cosine similarity over an unsupervised embedding, the neighborhood was fixed at 50 patients, and the neighbor pool was the training half of a single cohort. A supervised or metric-learned similarity, a different neighborhood size, a prevalence-recalibrated local estimate, or a neighbor pool drawn from several health systems could each perform differently, and none was tested. What the present analysis supports is narrower and still substantive. On this cohort, an unsupervised embedding whose geometry demonstrably tracks the outcome still yields a retrieval predictor that a fitted model beats decisively.

Absence of an advantage is not equivalence. No equivalence or noninferiority margin was prespecified. The correct statement is that this study found no evidence of superior internal discrimination for the embedding, bounded above by +0.019 ROC AUC, the upper

end of the paired interval. It is not that the two representations are equivalent.

What the embedding does offer is that it cost no discrimination, needed no explicit feature vector, and held across encoders. That is not a free lunch. It still depends on the hand-built deterministic narrative, which required the same parsing and structuring of the raw record. The engineering effort is relocated from designing a feature vector to designing a narrative template rather than eliminated.

Missing data were dropped rather than modeled. The three vital-sign columns were removed rather than imputed, because whether a vital sign was recorded is itself associated with the outcome. That rules out missingness completely at random without identifying the mechanism, so dropping the columns is conservative rather than justified. It costs three genuinely clinical predictors. It also discards signal the rest of the pipeline keeps: an absent categorical value becomes its own level in the feature vector, so a model can use the fact that a record is missing. Missingness in religion and race/ethnicity is likewise associated with the outcome, and Supplement M6 has the frequencies and the retained-category rules. The vitals are the one place this paper throws that signal away.

This study reports no fairness evidence. The ablation establishes only that neither model's predictions rely on the race/ethnicity and social-determinant fields supplied directly to it. Three questions remain open. Sociodemographic information may still reach the models indirectly through correlated psychiatric, medication, or utilization content. The outcome label is derived from prescribing behavior and may itself be inequitably distributed, given documented racial and ethnic disparities in the delivery of depression treatment [16,17]. And age, sex, preferred language, marital status, religion, and smoking status remain in both representations and were never permuted. The subgroup evidence does not return a uniform null: sex shows no difference anywhere, whereas the White-minus-non-White contrasts are directionally positive throughout without surviving correction, and non-White calibration is materially worse in the feature-vector arm.

This cohort cannot settle whether there is a race-associated gap, in either direction. The reason is sample size: the non-White stratum carries 302 TRD events against 1,181, so every contrast involving it has an interval about twice as wide. Individual race categories carry too few events to estimate at all, which is why the analysis compares White against everyone else. The groups where a disparity would be most likely to appear are the ones that disappear into that single comparison.

The cohort is one community health system. It skews middle-

aged and older (34.7% aged 65 or over), predominantly female (72.5%), overwhelmingly White (80.0%) and English-preferring (98.9%). Both the single-center design and that demographic profile limit transfer to more diverse or non-English-preferring populations and to health systems whose case mix or prescribing culture differs. Because the index prescription can be written anywhere in the system rather than in specialty psychiatric care, the cohort is also weighted toward routine rather than referred management.

The ablation perturbs one concept at a time. It therefore cannot disentangle co-correlated clinical content without joint perturbation, and prior treatment exposure and medication burden are the two concepts most likely to share contribution. It also cannot rule out reconstruction of a permuted concept from correlated fields left intact.

The operating point and the calibration are both optimistic. The Youden-J threshold was selected on the same held-out patients the models were then scored on, so the reported sensitivity and specificity are not candidate operating characteristics. Calibration was measured inside a case-enriched sample. Its 17.5% TRD rate was chosen by the sampling design and is not what a clinic would see, so a predicted risk from these models is a risk relative to this cohort. Applying them anywhere else would need the probabilities rescaled to that population's own rate.

Implications and future directions

At 0.65, none of this is ready for the clinic. The next study should validate externally, across sites and forward in time, with the whole pipeline frozen, and should report calibration and decision-curve analysis rather than discrimination alone. This cohort was case-enriched, so absolute risks need rescaling before they mean anything clinically.

Incremental value should be tested against simple baselines, not only against other architectures. The comparators that matter are outcome prevalence, age and coded severity, prior medication history, and a compact set of comorbidity and utilization variables. An embedding earns its place only if it improves transportability, calibration or net benefit beyond those.

Three experiments follow from what is reported here. Embed an un-tailored record, which tests whether an encoder finds TRD-relevant structure without being told where to look. The obstacle there is input length: predictors here come from a fixed 730-day window, whereas recorded history reaches a median of 1,792 days and a maximum of 5,288 across a median of 21 encounters. Every way around that

reintroduces a design decision, which is itself worth reporting. Learn a similarity against the outcome, or draw neighbors from several health systems, and see whether retrieval closes the 0.063 gap. And validate the target itself, by chart adjudication and symptom measurement, to find out how often a treatment switch means a treatment failed. Until that is done, the defensible use is research-stage cohort enrichment, not treatment selection.

One thing retrieval offers that a fitted model does not is that its neighbors are real patients with observed courses. Whether that is worth 0.063 ROC AUC is a question about clinical workflow rather than discrimination, and it needs a different study. The feature vector's explicit treatment-exposure variables also make a different question tractable: which patients benefit from which next step, rather than who becomes resistant. That is ongoing work.

Conclusions

Writing a structured EHR out as text and embedding it with a general-purpose transformer extracted the same predictive signal as a typed feature vector built from the same record, and no more of it. The embedding's contribution was reorganization rather than discrimination, and its apparent advantage or disadvantage depended entirely on which classifier was applied to it.

Predicting a patient from the recorded outcomes of their nearest neighbors recovered part of that signal and still lost decisively to a model fitted on the same embedding, on every encoder tested. The digital-twin premise in its unsupervised geometric form is therefore not competitive here.

Both results rest on hand-picked patient data. Predictor selection ran once and served both arms, so the engineering was relocated rather than removed, and neither result is evidence about what a model would do with a raw record. Embedding a whole untailored history was not attempted because record length stands in the way, and that experiment remains the interesting one.

Nothing here is close to clinical utility at a ROC AUC near 0.65, and the outcome is a treatment-switch pattern whose relationship to verified non-response is untested. Three things would decide whether any of it matters. The first is whether the signal survives in another health system. The second is whether it beats a simple baseline built from care history alone. The third is whether it finds patients whose symptoms were measured and did not improve, rather than patients whose path through the system was shaped by insurance, access and prescribing habit.

Declarations

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Conflicts of interest. None declared.

Ethics and data handling. This study is a secondary analysis of de-identified data and is therefore not human-subjects research subject to institutional review board approval; no IRB review, protocol number, or waiver of informed consent applies. The analysis used a de-identified EHR extract containing no direct patient identifiers, and no protected health information was accessible to the investigators at any stage. All computation was performed on local institutional hardware, including the sentence-transformer embedding of the patient narratives. No patient data, structured or narrative, was transmitted to any external, third-party, or commercial service or application programming interface.

Authors' contributions. MF designed and implemented the analysis pipeline, performed all modelling and statistical analysis, produced the figures and tables, and wrote the manuscript. MP directed the project and supervised its scientific design and interpretation. KLF prepared the upstream clinical data, including the treatment-resistant depression labels and the antidepressant index dates that define the cohort and the outcome. RK, SS, and DP provided methodological feedback and guidance on the analysis and its interpretation throughout. All authors reviewed and approved the final manuscript.

Patient and public involvement. No patients or members of the public were involved in the design, conduct, reporting, or dissemination of this study.

Protocol. A study protocol was not prepared.

Registration. This study was not registered.

Data availability. The EHR data are not publicly shareable. The analysis code is publicly available [29]. Fitted model objects are not distributed, as they are derived from non-shareable patient data.

Reporting. This study is reported in accordance with the TRIPOD+AI guideline for prediction-model studies [3]. The completed TRIPOD+AI checklist is provided as a separate submission document.

Abbreviations

AUPRC: area under the precision-recall curve.

CI: confidence interval.

EHR: electronic health record.

EMBEDDED: the embedded representation, a deterministic patient narrative encoded. by a pretrained sentence-transformer

EPV: events per variable.

FEATURE: the typed feature representation, a typed vector built directly from. coded EHR fields

IQR: interquartile range.

MDD: major depressive disorder.

ROC AUC: area under the receiver operating characteristic curve.

SDOH: social determinants of health.

SMD: standardized mean difference.

TRD: treatment-resistant depression.

TRIPOD+AI: Transparent Reporting of a multivariable prediction model for. Individual Prognosis Or Diagnosis, artificial intelligence extension

WCE: weighted calibration error.

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Multimedia Appendix 1. FEATURE predictor inventory

The FEATURE representation comprises 59 source fields (the three within-patient mean vital signs having been dropped at load time; see *Missing data and sample partition*). Quantitative and boolean fields map one-to-one to model columns. The eight categorical fields are one-hot encoded (drop='if_binary' for the single binary field, all

levels retained otherwise), and five of them carry an additional automatic “Missing” level for patients with no recorded value. The resulting design matrix has 92 columns (15 quantitative + 36 boolean + 41 one-hot levels). Display names are those used in the feature-importance figures.

Table A1. Quantitative predictors (15; continuous, standardized).

Predictor
MDD history before index (days)
History length (days)
Encounter count
ED visits (count)
Inpatient days before index
Age (years)
Medications active at index
Anti-inflammatories active at index
Sleep meds active at index
Anxiolytic days before index
Bupropion trials (6+ wk)
Mirtazapine trials (6+ wk)
SNRI trials (6+ wk)
SSRI trials (6+ wk)
Vortioxetine trials (6+ wk)

Table A2. Boolean predictors (36; cast to 0/1), grouped by block.

Block	Predictors
Clinical flags (3)	MDD in history; Suicidality; Augmentation therapy
Psychiatric comorbidities (8)	Adjustment disorder; Anxiety disorder; Dysthymia (chronic depression); Insomnia; OCD; PTSD; Social anxiety disorder; Substance use disorder (any)
Medical comorbidities (4)	Chronic pain; Diabetes; High cholesterol; Thyroid disorder
Prescribing-constraint / safety (2)	Seizure disorder; Uncontrolled hypertension
Substance use disorders (10)	Alcohol; Cannabis; Cocaine; Hallucinogen; Inhalant; Nicotine; Opioid; Other stimulant; Other substance; Sedative/hypnotic

Block	Predictors
Social determinants of health (9)	Education or literacy issue; Employment issue; Housing or financial issue; Legal or criminal issue; Occupational hazard exposure; Family/support group issue; Psychosocial circumstances; Social environment issue; Upbringing issue

Table A3. Categorical predictors (8 fields → 41 one-hot columns). “Missing” indicates an automatic one-hot level for unrecorded values.

Field	One-hot columns	Levels
Sex	1	Male (Female = reference, dropped)
Preferred language	6	Asian/Pacific Islander; English; Other; Other Indo-European; Spanish; Missing
Marital status	6	Divorced; Never married; Married; Separated; Widowed; Missing
Religion	6	Catholic; Non-Christian; Orthodox; Other/Unknown; Protestant; Missing
Smoking status	4	Current; Former; Never; Missing

Field	One-hot columns	Levels
Race/ethnicity	8	American Indian/Alaska Native; Asian; Black/African American; Hispanic/Latino; Multi-race; Native Hawaiian/Pacific Islander; White/Caucasian; Missing
MDD recurrence	4	Unspecified; Single episode; Recurrent; Dysthymia (chronic depression)
MDD severity	6	Unspecified; Mild; Moderate; Severe; Psychotic; Remission